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Mallik et al.

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(54) **ULTRA-THIN, HYPER-DENSITY SEMICONDUCTOR PACKAGES**

(71) Applicant: **Intel Corporation**, Santa Clara, CA (US)

(72) Inventors: **Debendra Mallik**, Chandler, AZ (US); **Robert L. Sankman**, Phoenix, AZ (US); **Robert Nickerson**, Chandler, AZ (US); **Mitul Modi**, Phoenix, AZ (US); **Sanka Ganesan**, Chandler, AZ (US); **Rajasekaran Swaminathan**, Chandler, AZ (US); **Omkar Karhade**, Chandler, AZ (US); **Shawna M. Liff**, Scottsdale, AZ (US); **Amruthavalli Alur**, Tempe, AZ (US); **Sri Chaitra J. Chavali**, Chandler, AZ (US)

(73) Assignee: **Intel Corporation**, Santa Clara, CA (US)

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H01L 23/00 (2006.01)
(Continued)

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CPC **H01L 23/49811** (2013.01); **H01L 23/3128** (2013.01); **H01L 23/5389** (2013.01); **H01L 24/29** (2013.01)

(58) **Field of Classification Search**
CPC H01L 23/49811; H01L 23/3128; H01L 23/5389; H01L 24/29
See application file for complete search history.

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Primary Examiner — Fernando L Toledo

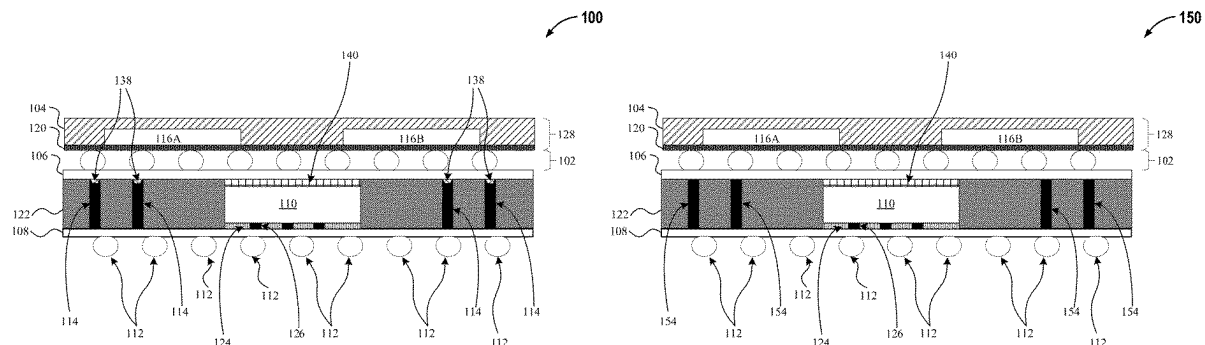
Assistant Examiner — Neil R Prasad

(74) *Attorney, Agent, or Firm* — Schwabe, Williamson & Wyatt P.C.

(57) **ABSTRACT**

Ultra-thin, hyper-density semiconductor packages and techniques of forming such packages are described. An exemplary semiconductor package is formed with one or more of: (i) metal pillars having an ultra-fine pitch (e.g., a pitch that is greater than or equal to 150 μm , etc.); (ii) a large die-to-package ratio (e.g., a ratio that is equal to or greater than 0.85, etc.); and (iii) a thin pitch translation interposer. Another exemplary semiconductor package is formed using coreless substrate technology, die back metallization, and low temperature solder technology for ball grid array (BGA) metallurgy. Other embodiments are described.

20 Claims, 15 Drawing Sheets



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(51) **Int. Cl.**

H01L 23/31 (2006.01)

H01L 23/538 (2006.01)

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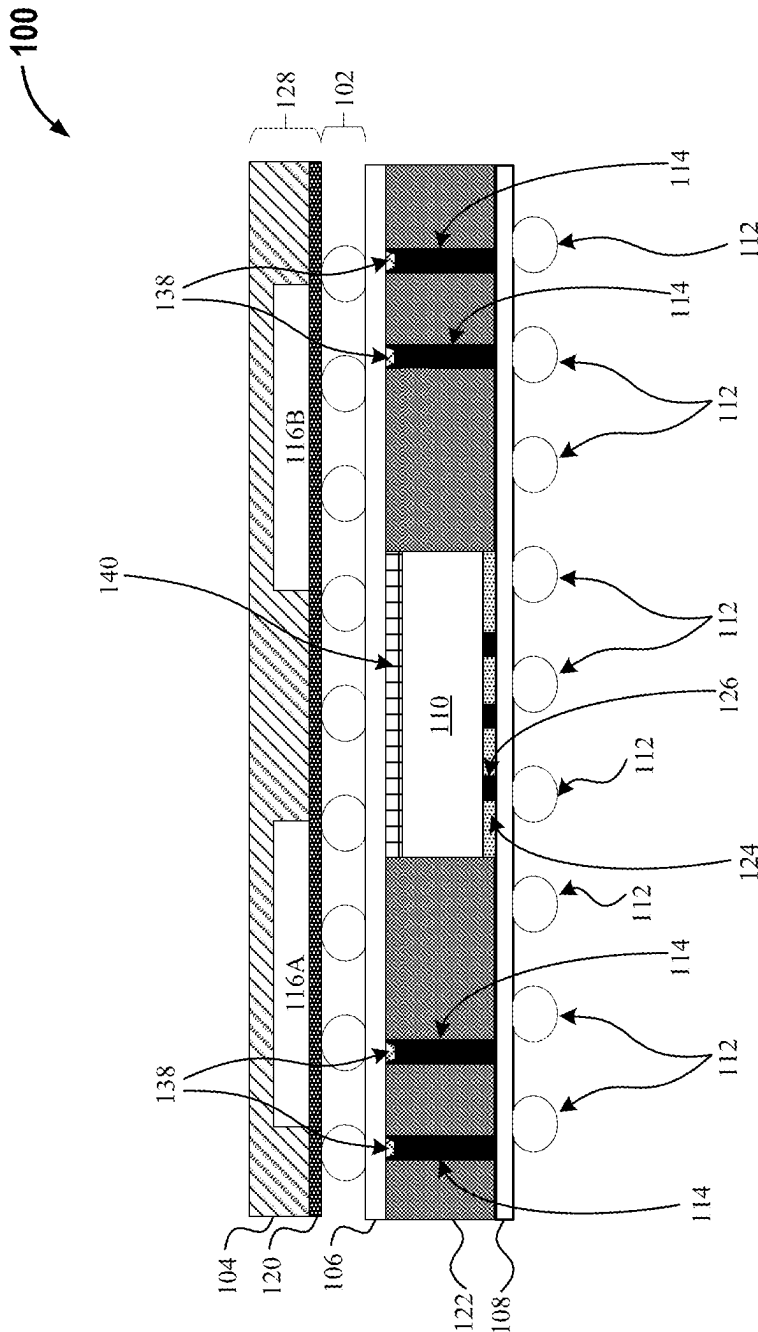


FIG. 1A

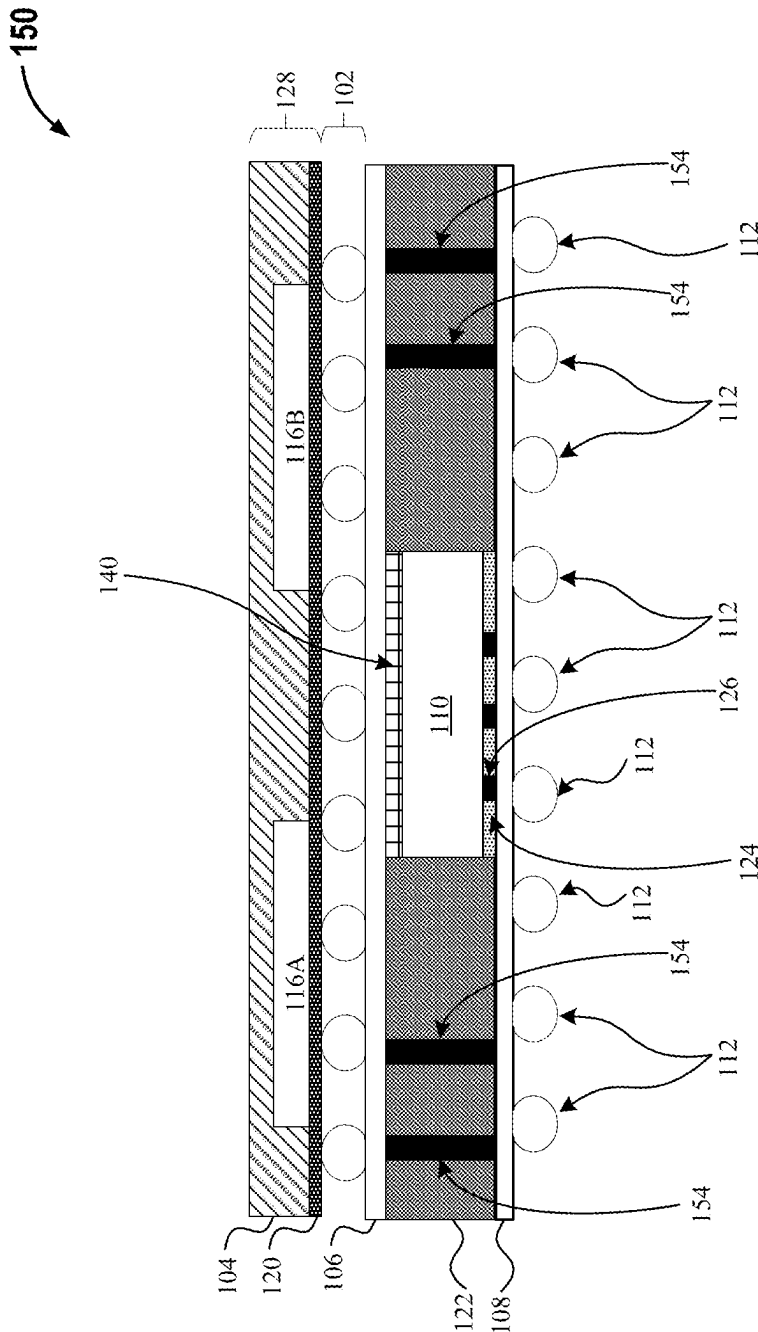


FIG. 1B

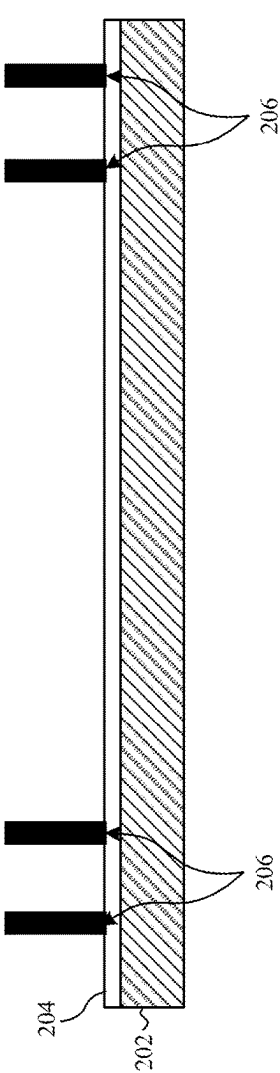


FIG. 2A

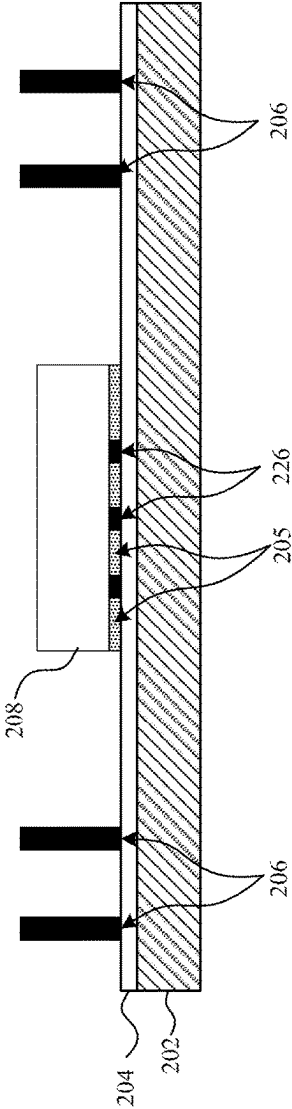


FIG. 2B

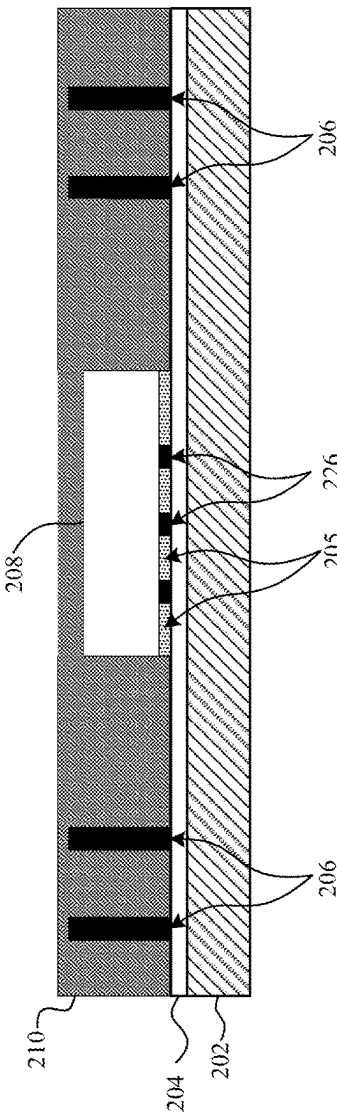


FIG. 2C

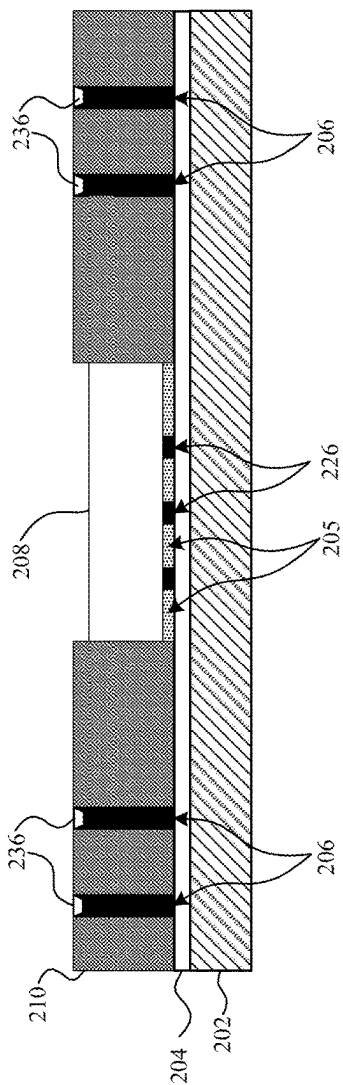


FIG. 2D

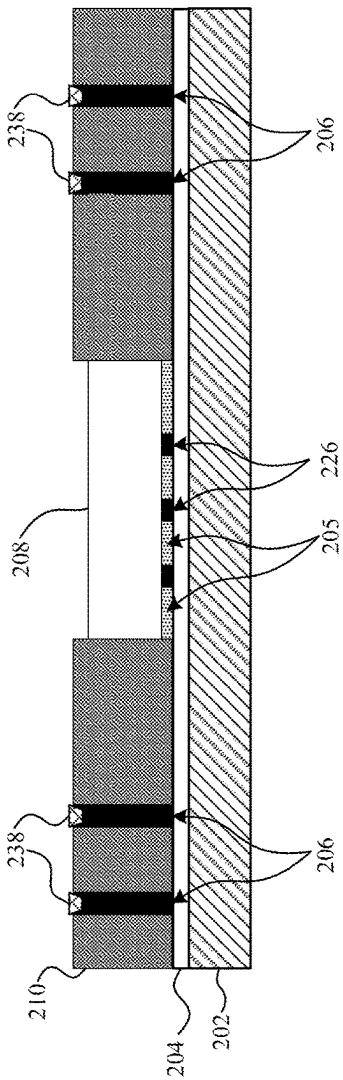


FIG. 2E

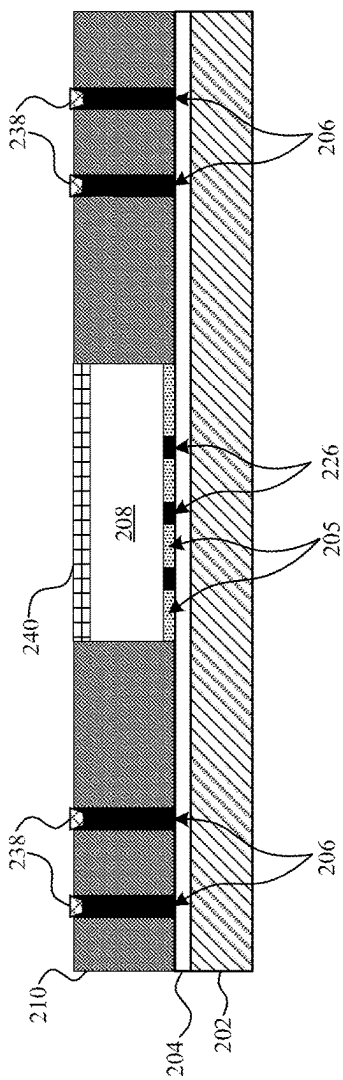


FIG. 2F(i)

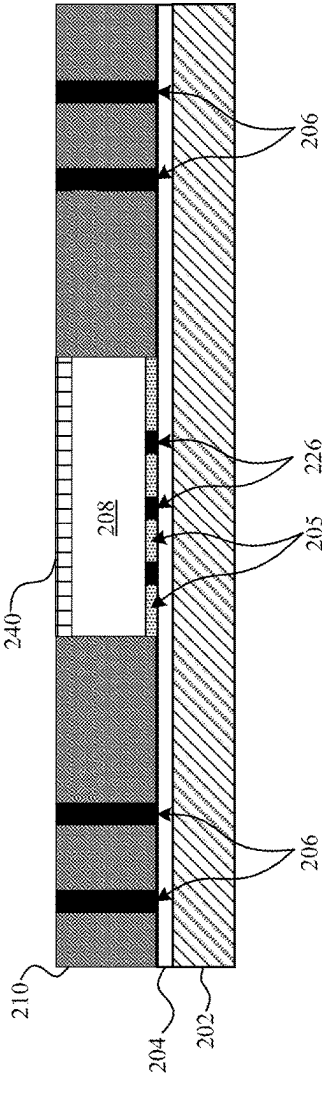


FIG. 2F(ii)

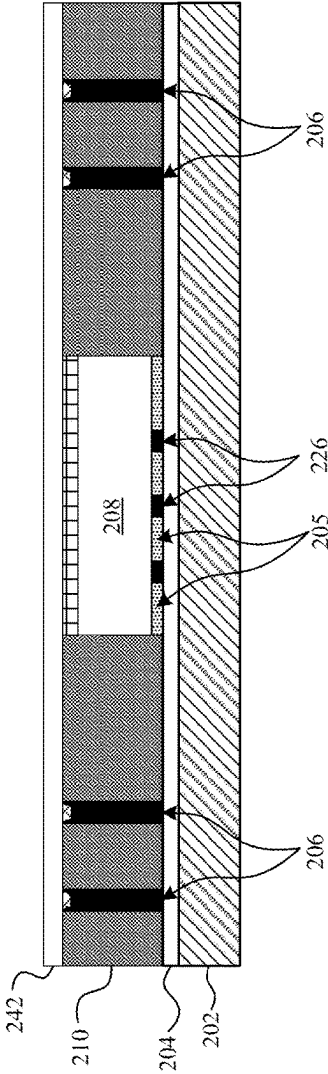


FIG. 2G

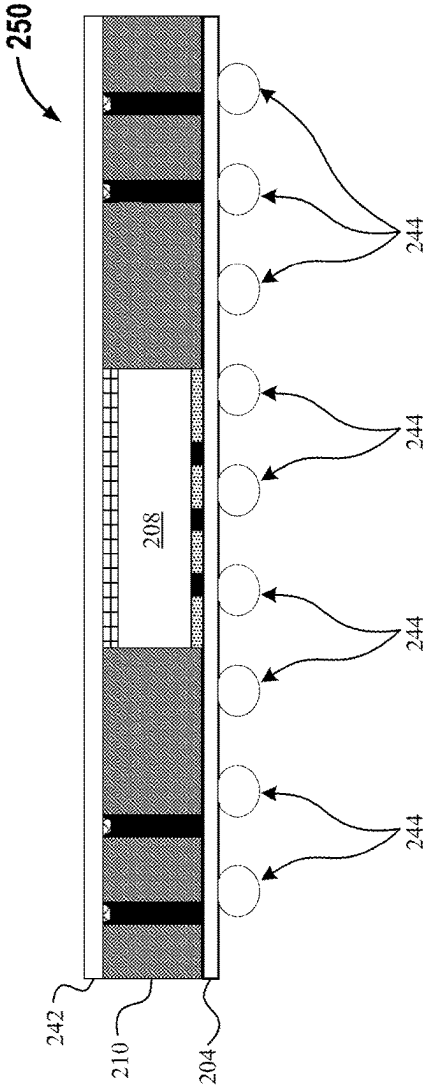


FIG. 2H

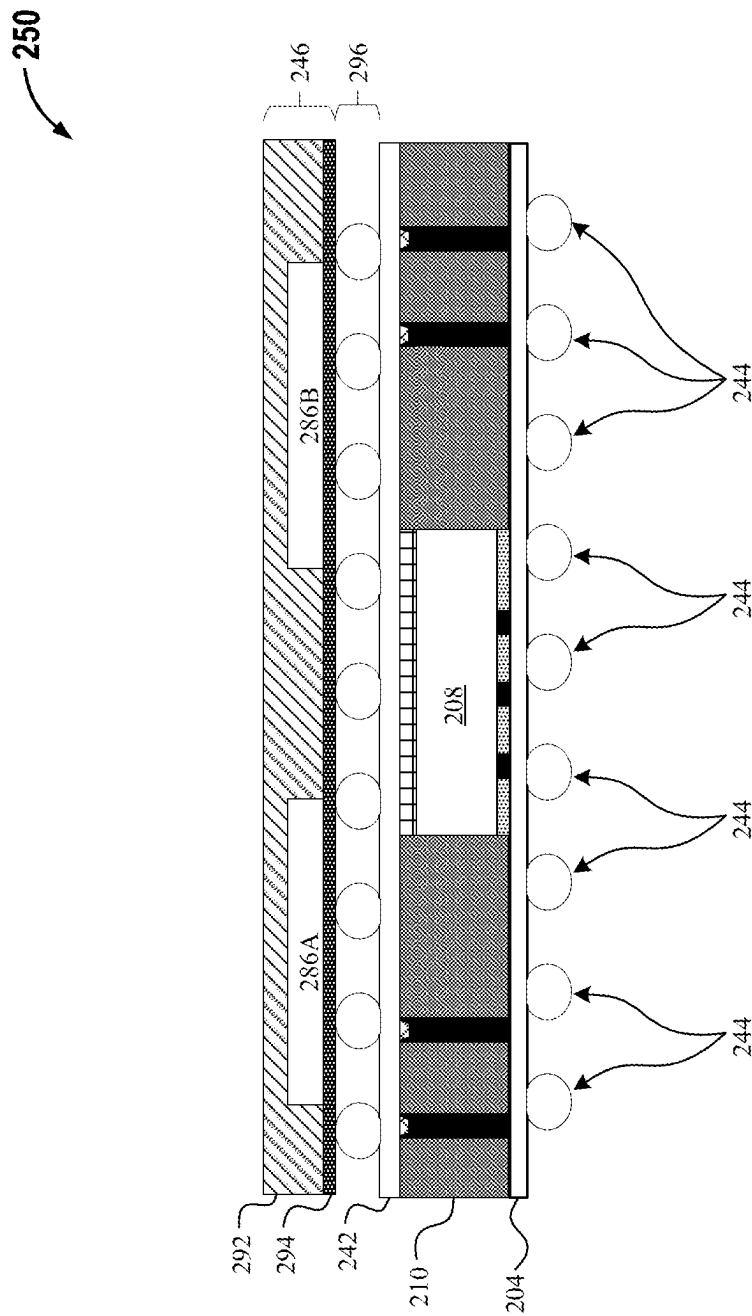


FIG. 2I

300

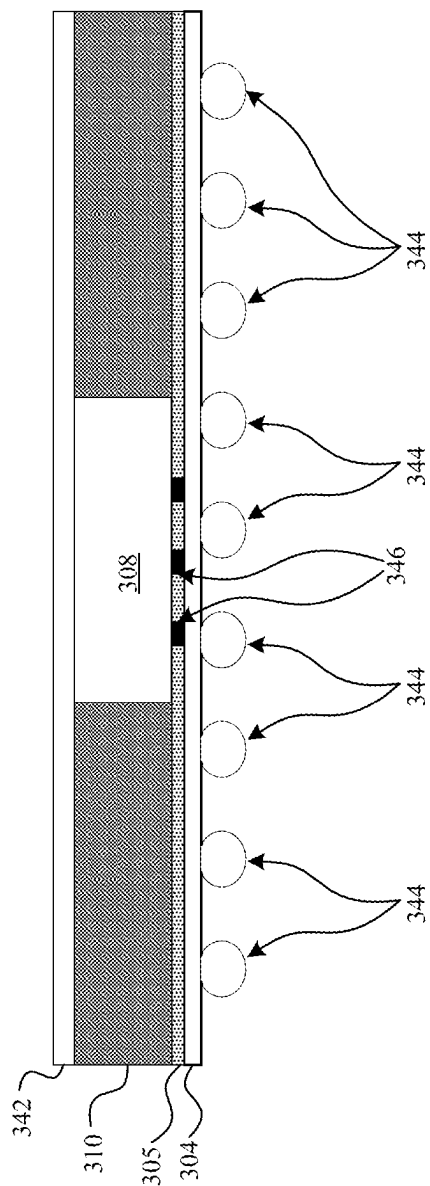


FIG. 3

400

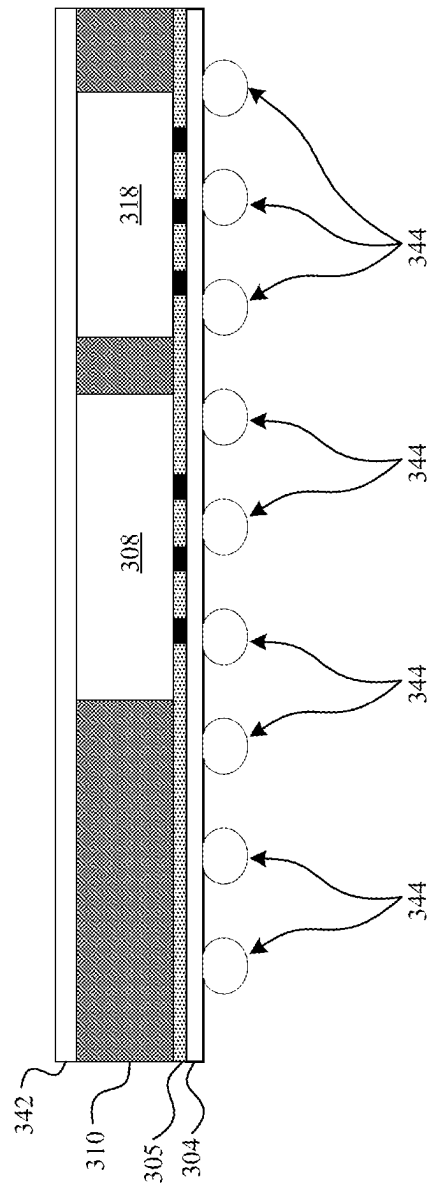


FIG. 4

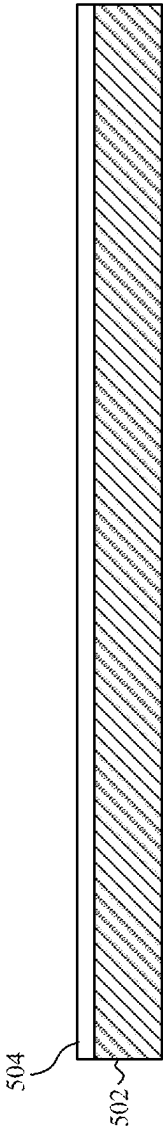


FIG. 5A

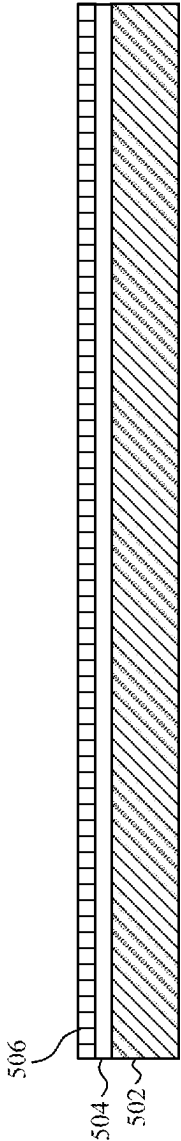


FIG. 5B

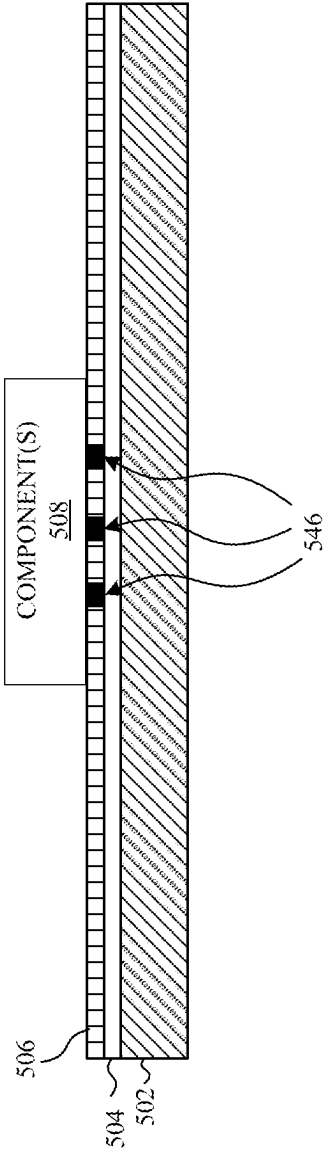


FIG. 5C

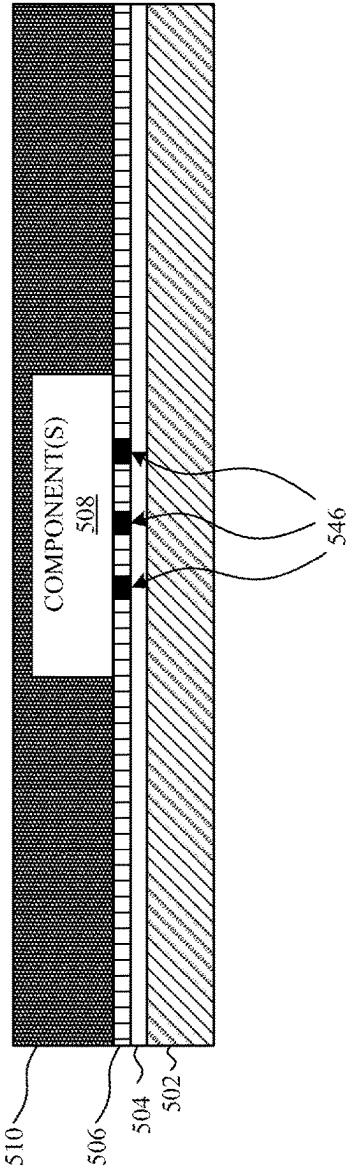


FIG. 5D

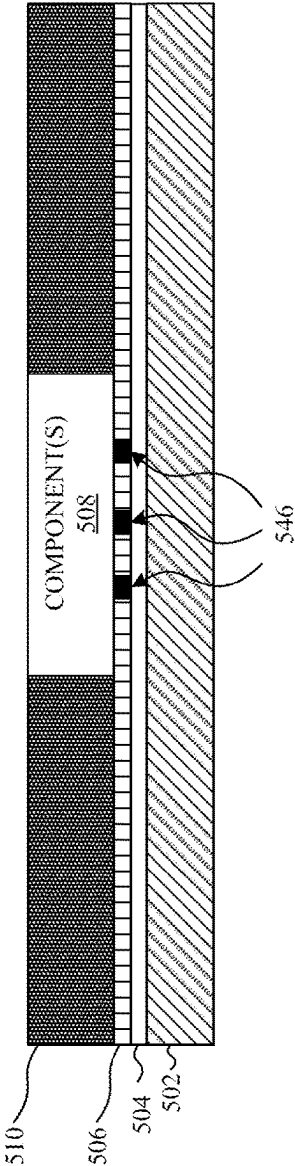


FIG. 5E

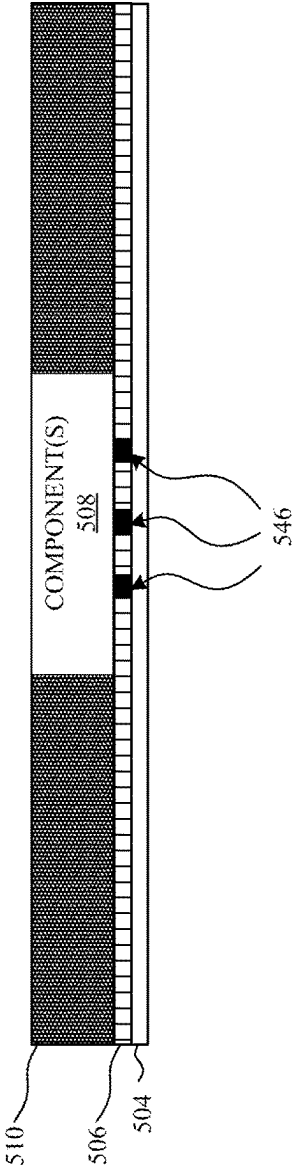


FIG. 5F

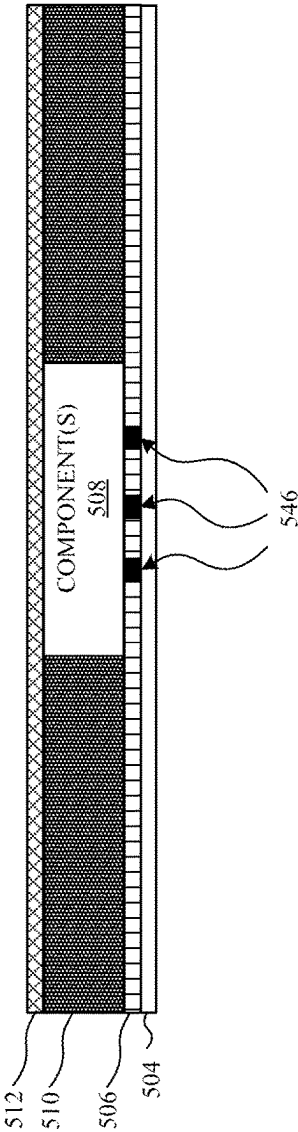


FIG. 5G

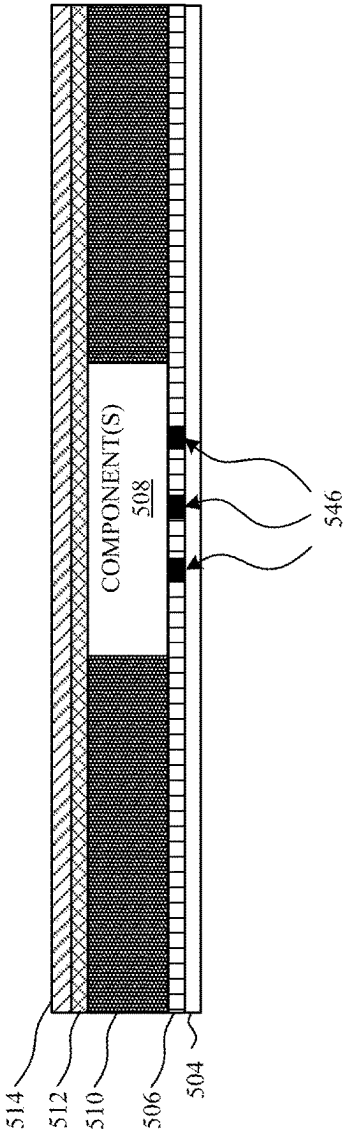


FIG. 5H

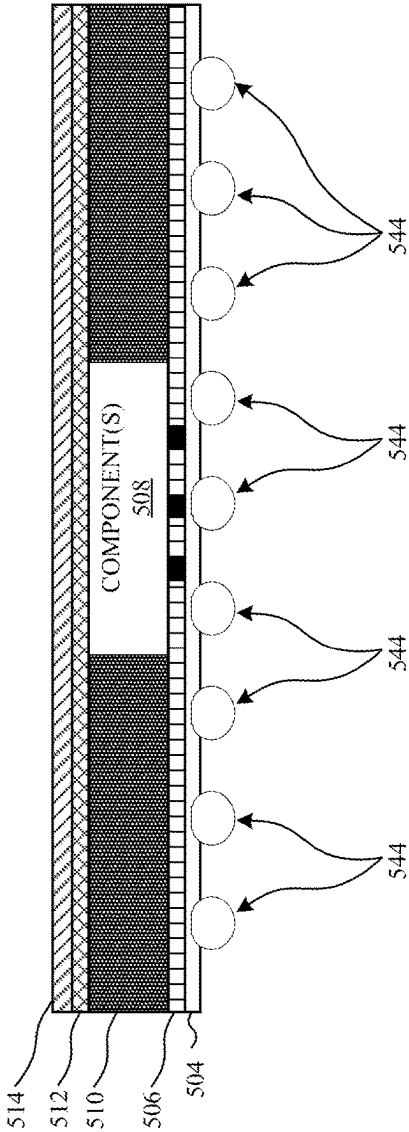


FIG. 5I

575

600

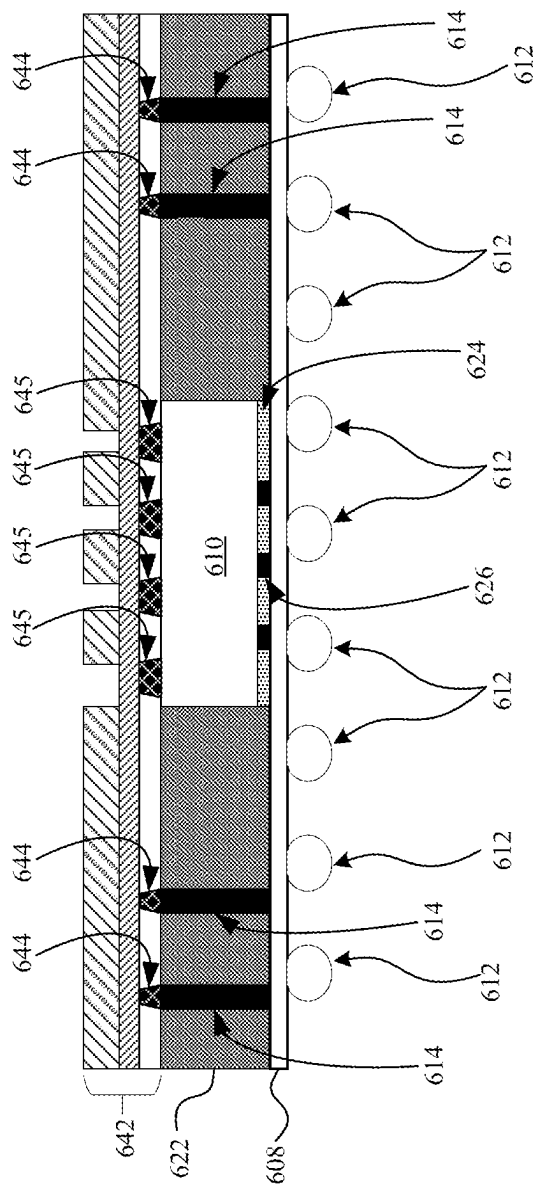


FIG. 6

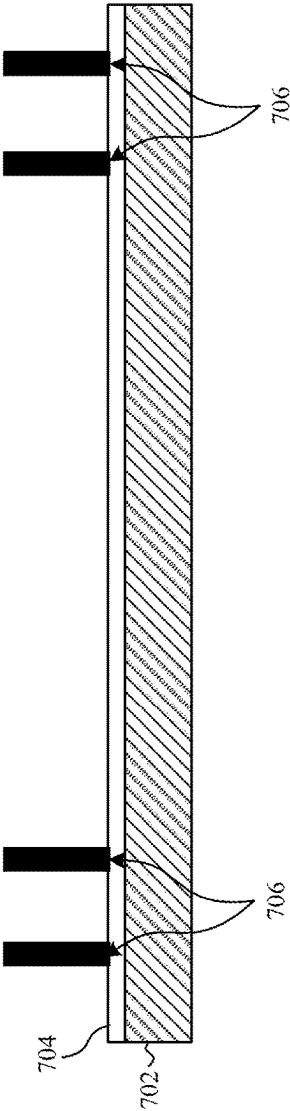


FIG. 7A

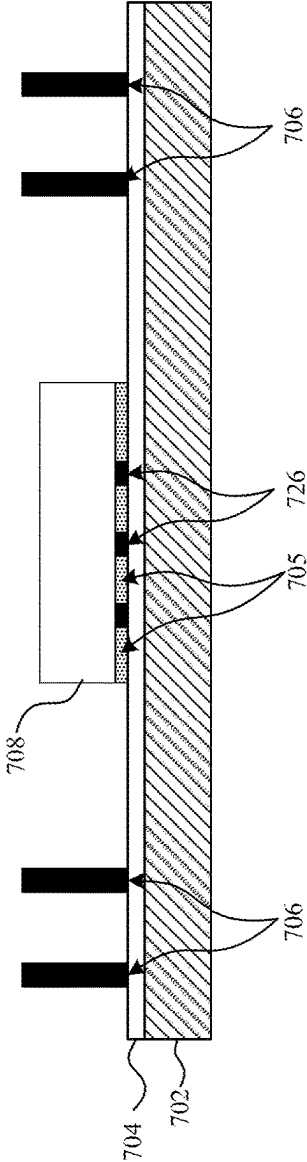


FIG. 7B

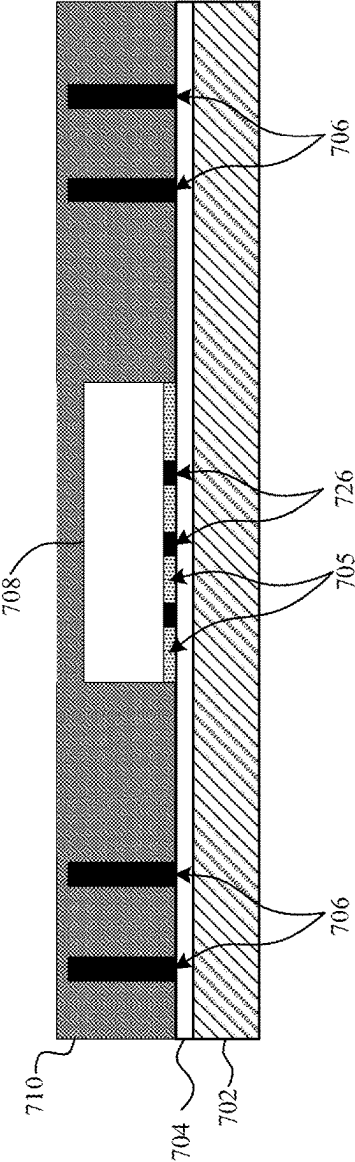


FIG. 7C

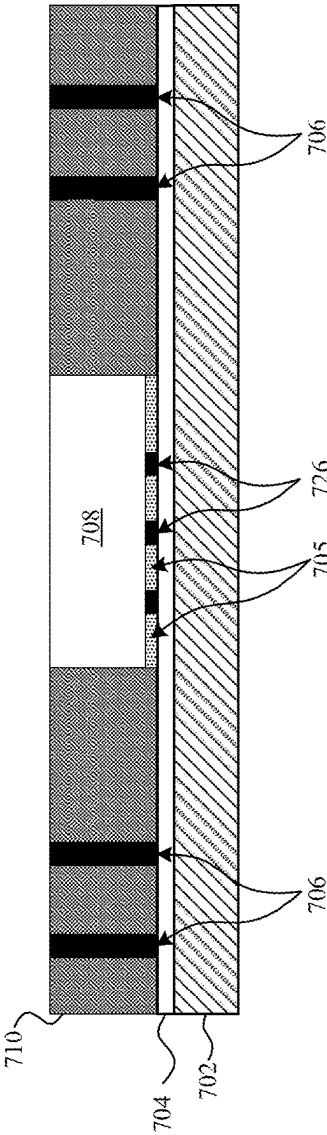


FIG. 7D

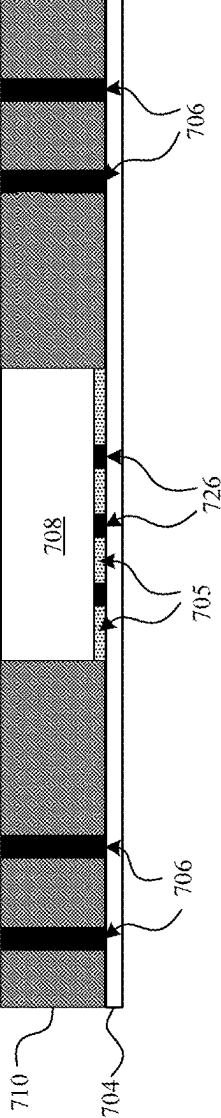


FIG. 7E

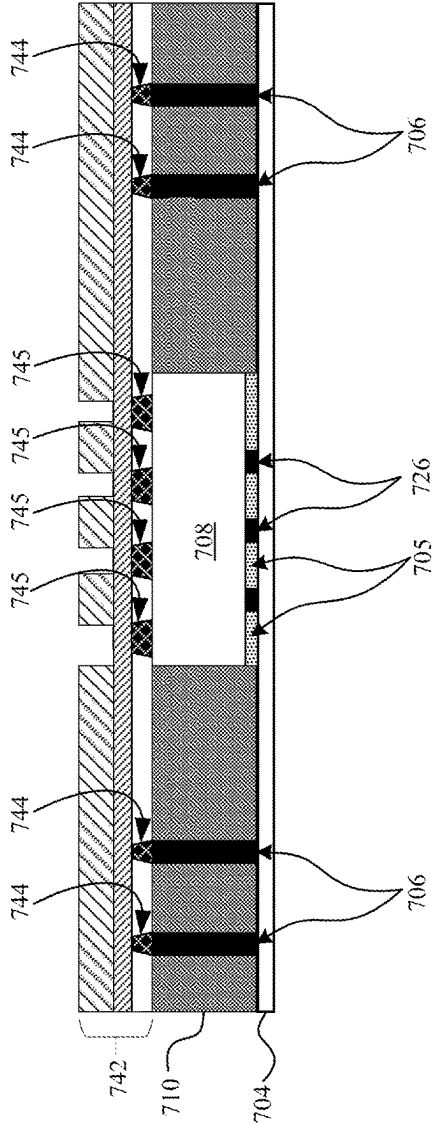


FIG. 7F

700

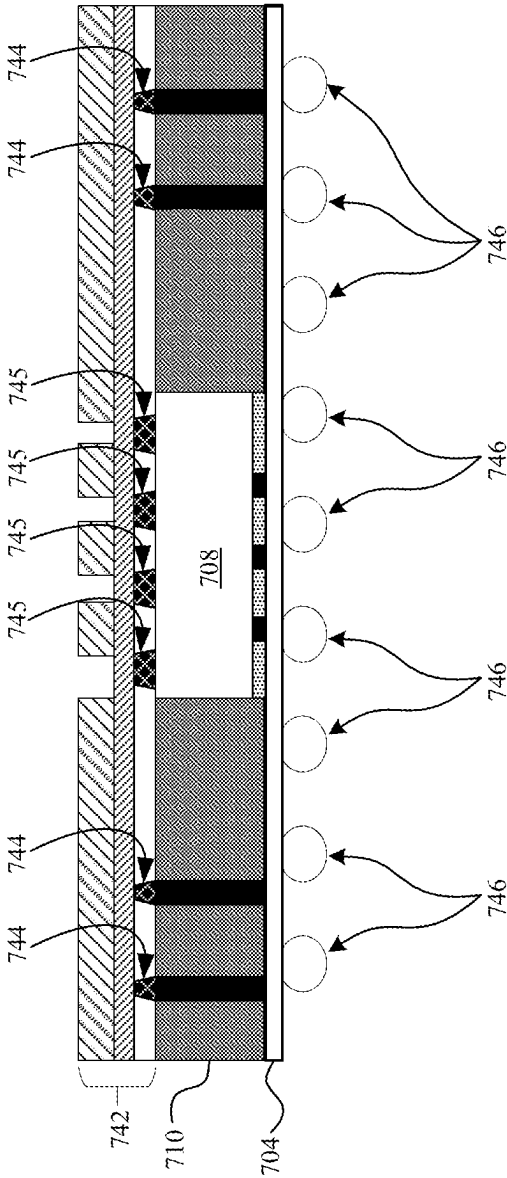


FIG. 7G

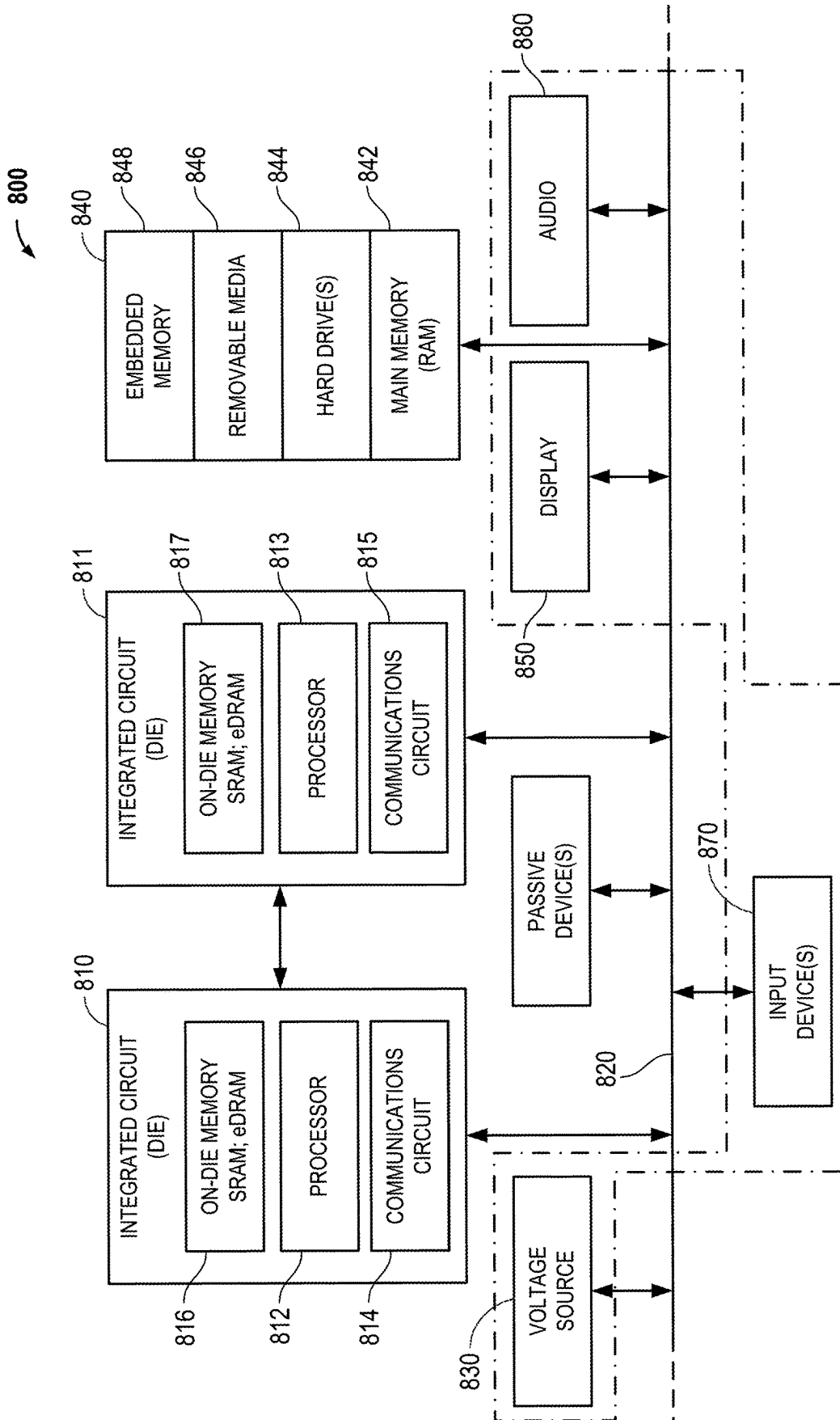


FIG. 8

1

**ULTRA-THIN, HYPER-DENSITY
SEMICONDUCTOR PACKAGES****CROSS-REFERENCE TO RELATED
APPLICATION**

This patent application is a continuation of U.S. patent application Ser. No. 17/862,300, filed Jul. 11, 2022, which is a continuation of U.S. patent application Ser. No. 16/646,529, filed Mar. 11, 2020, now U.S. Pat. No. 11,430,724, issued Aug. 30, 2022, which is a U.S. National Phase Application under 35 U.S.C. § 371 of International Application No. PCT/US2017/069138, filed Dec. 30, 2017, entitled “ULTRA-THIN, HYPER-DENSITY SEMICONDUCTOR PACKAGES,” which designates the United States of America, the entire disclosure of which are hereby incorporated by reference in their entirety and for all purposes.

FIELD

Embodiments generally relate to semiconductor packages. More specifically, embodiments relate to ultra-thin, hyper-density semiconductor packages and techniques of forming such packages.

BACKGROUND INFORMATION

Conventional semiconductor package substrates typically include at least one core layer impregnated in a dielectric material to provide mechanical rigidity to the substrate. Latest trends of electronic devices such as mobile phones, mobile internet devices (MIDs), multimedia devices and computer notebooks demand for slimmer and lighter designs. Coreless substrates are adopted for fabrication of components in such electronic devices to enable a thinner profile of the components. The thickness of coreless substrates can be, for example, as little as approximately 25% of the thickness of cored substrates.

Cored and coreless substrates may be susceptible to warpage problems during Surface Mount Technology (SMT) processes. Furthermore, coreless substrates, in some scenarios, may be more susceptible to warpage problems during SMT processes (when compared to conventional substrates with core layers). SMT processes typically involve subjecting package substrates to heating and cooling which in turn create expansion and contraction of the substrate. The difference in coefficient of thermal expansion (CTE) of the various materials forming the substrate results in different rates of expansion and contraction and hence stress in the substrate. The resulting stress warps the substrate and causes manufacturing problems during component package assembly as well as during performance of SMT processes. As demand for smaller, and higher performing devices continues to grow, packages will get thinner and pitch (e.g., spacing between package components, etc.) will get finer, which may increase the occurrence of warpage in cored or coreless packages. Increased warpage can undesirably result in failure or reduced performance of packages or increase problems related to the reliability of electronic devices having packages therein.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments described herein are illustrated by way of example and not limitation in the figures of the accompanying drawings, in which like references indicate similar

2

features. Furthermore, in the figures, some conventional details have been omitted so as not to obscure from the inventive concepts described herein.

FIGS. 1A-1B are cross-sectional illustrations of ultra-thin, hyper-density semiconductor packages according one or more embodiments.

FIGS. 2A-2I are cross-sectional side view illustrations of a method of forming an ultra-thin, hyper-density semiconductor package according an embodiment.

FIG. 3 is a cross-sectional illustration of an ultra-thin, hyper-density semiconductor package according another embodiment.

FIG. 4 is a cross-sectional illustration of an ultra-thin, hyper-density semiconductor package according yet another embodiment.

FIGS. 5A-5I are cross-sectional side view illustrations of a method of forming an ultra-thin, hyper-density semiconductor package according another embodiment.

FIG. 6 is a cross-sectional illustration of an ultra-thin, hyper-density semiconductor package according one or more embodiments.

FIGS. 7A-7G are cross-sectional side view illustrations of a method of forming an ultra-thin, hyper-density semiconductor package according one or more embodiments.

FIG. 8 is an illustration of a schematic block diagram of a computer system that utilizes an ultra-thin, hyper-density semiconductor package, according to an embodiment.

DETAILED DESCRIPTION

Embodiments described herein provide ultra-thin, hyper-density semiconductor packages and techniques of forming such packages. One advantage of the ultra-thin, hyper-density semiconductor packages fabricated in accord with the embodiments described herein is that such packages suffer from minimal or no warpage (when compared to cored and/or coreless packages fabricated using conventional techniques). In this way, packages fabricated in accord with the embodiments described herein can assist with avoiding warpage problems that occur during surface mount technology (SMT) processes. Furthermore, the embodiments described herein can assist with fabrication of packages having: (i) an ultra-thin z-height (e.g., a z-height that is less than or equal to 1 mm, etc.); and (ii) a die-to-package ratio (e.g., a ratio that is equal to or greater than 0.7, etc.). Such packages can be used in handheld and mobile-client products.

For one embodiment, a semiconductor package is formed with: (i) metal pillars having an ultra-fine pitch (e.g., a pitch that is greater than or equal to 150 μm , etc.); (ii) a large die-to-package ratio (e.g., a ratio that is equal to or greater than 0.85, etc.); and (iii) a thin pitch translation interposer. For one embodiment, a semiconductor package is formed using coreless substrate technology, die back metallization, and low temperature solder technology for ball grid array (BGA) metallurgy. Other embodiments are described below in connection with one or more of FIGS. 1A-8.

FIG. 1A is a cross-sectional illustration of an ultra-thin, hyper-density semiconductor package **100** that includes a package-on-package (PoP) architecture according one or more embodiments. The package **100** includes solder bumps **112**, which may be low temperature solder bumps formed from tin (Sn) or tin alloys (e.g., Sn—Al alloys, Sn—In alloys, Sn—Bi alloys, etc.). For one embodiment, the solder bumps **112** are formed from an Sn-57 Bi-1 alloy.

The package **100** also includes a high density (HDP) ultra-thin substrate **108** onto which the solder bumps **112** are

formed. The HDP substrate **108** may be formed from any suitable material (silicon, glass, metal, etc.). For one embodiment, the substrate **108** has a nominal thickness (i.e., z-height) of approximately 66 μm . For one embodiment, a top side of the substrate **108** has metal pillars **114** (e.g., copper pillars, etc.) formed thereon. For one embodiment, the metal pillars **114** have a maximum nominal thickness (i.e., z-height) that is approximately 150 μm . In a specific embodiment, a low temperature solder material (e.g., Sn57 Bi, SAC305, etc.) may be inserted into shallow holes formed in the top surfaces of the pillars **114** and reflowed to form solder caps **138**. For one embodiment, the caps **138** are above the mold compound **122** by a predetermined z-height (e.g., approximately 10 μm , etc.). For one embodiment, low temperature solder may be inserted or applied to the shallow holes via a paste print solder process that involves using a stencil or via an injection molded solder (IMS) process. The solder caps **138** may be planarized and cleaned to achieve a desired z-height.

The package **100** also includes a component **110**. For one embodiment, the component **110** can include one or more of a system-on-chip (SoC), a central processing unit (CPU) component, a memory, a processor, a Platform Controller Hub (PCH), a Peripheral Component Interconnect (PCI), a Graphics Processing Unit (GPU), an on-chip system fabric, a network interface controller, a stacked component, a non-stacked component, a ball grid array (BGA) package, any other electronic component, or any combination thereof. The component **110** may include one or more semiconductor dies mounted on the HDP substrate **108**. The dies of the component **110** can be attached to the HDP substrate **108** according to a variety of suitable configurations including, a flip-chip configuration or other configurations such as wire bonding and the like. In the flip-chip configuration, an active side of the dies **102** is attached to a surface of the substrate **108** using interconnect structures such as bumps or pillars. Examples of such interconnect structures include, but are not limited to, Cu bumps, any type of low-lead or lead-free solder bumps, tin-copper bumps, Cu pillars, combinations thereof, or the like. The active side of the dies in the component **110** may have one or more transistor devices formed thereon. Each of the dies in the component **110** may represent a discrete chip. The dies in the component **110** may, include, or be a part of a processor, memory, or application specific integrated circuit (ASIC).

As shown in FIG. 1A, the component **110** may be coupled to the substrate **108** via the first level interconnects (FLIs) **126** and an epoxy layer **124**. The FLIs **126** can have a nominal thickness (i.e., a z-height, etc.) of approximately 35 μm . The epoxy layer **124** can have a nominal thickness (i.e., a z-height, etc.) of approximately 25 μm . For some embodiments, a substrate pad may have nominal thickness (i.e., a z-height, etc.) of 40 μm . For some embodiments, a pad-trace component may have nominal thickness (i.e., a z-height, etc.) of 10 μm .

The component **110** and the pillars **114** may be encapsulated in a first mold compound **122**. The component **110** may be a monolithic package (e.g., a monolithic SoC, etc.). Furthermore, the component **110** may be designed to have a nominal thickness (i.e., z-height) of approximately 125 μm .

The package **100** also includes an epoxy material **140** with a predetermined thickness (e.g., approximately 25 μm , etc.) and a predetermined thermal conductivity (e.g., approximately 3-5 W/mK, etc.) applied on the exposed top surface of the component **110**. The epoxy material **140** can be a paste or a film. When the epoxy material **140** is a paste,

it is printed onto the component **110**. When the epoxy material **140** is a film, it is laminated onto the component **110**.

For one embodiment, the pillars **114** (e.g., the solder caps **138**, etc.) couple the substrate **108** to a pitch translation interposer **106**. This coupling may be performed by reflow of the solder caps **138**. Furthermore, the component **110** may be attached to the pitch translation interposer **106** to enable connections between the component **110** and another package **128**. For one embodiment, the pitch translation interposer **106** has a nominal thickness (i.e., z-height) of approximately 60-63 μm . Interconnect structures **102** may be used for coupling the interposer **106** to the package **128**. For one embodiment, the interconnect structures **102** and the pitch translation interposer **106** collectively have a nominal thickness (i.e., z-height) of approximately 63 μm .

As alluded to above, the package **100** also includes a package **128** coupled to the component **110** via the interposer **106**. The package **128** may one or more components **116A-B** (e.g., one or more semiconductor dies, a system-on-chip (SoC), a central processing unit (CPU) component, a memory, a processor, a Platform Controller Hub (PCH), a Peripheral Component Interconnect (PCI), a Graphics Processing Unit (GPU), an on-chip system fabric, a network interface controller, a stacked component, a non-stacked component, a ball grid array (BGA) package, any other electronic component, or any combination thereof, etc.). The package **128** may also comprise one or more layers **120** (e.g., dielectric layers, metal layers, other layers, etc.), and electrical connections (not shown). These electrical connections include, but are not limited to, wire bonds. The package **128** may also comprise a mold compound **104** that encapsulates the components **116A-B** and the one or more layers **120**. As used herein, "encapsulating" does not require all surfaces of the components **116A-B** to be encased within a mold compound. For a first example, the top surfaces of the layer(s) **120** are encased in the mold compound **120**, while the mold compound **120** is not formed over the lateral surfaces of the pillars **206**. For a second example, and as illustrated in FIG. 1A, the lateral and top sides of the components **116A-B** are encased in the mold compound **120**. Additional encapsulation operations may be subsequently performed in order to provide chemical and mechanical protection to the top surface of the package **128**. In some embodiments, the amount of mold compound **104** is controlled to achieve a specified z-height. Alternatively, an amount of the mold compound **104** can be removed after application in order to expose the top and/or lateral surfaces of the package **128**. As shown in FIG. 1A, it is not required that the top surfaces of the package **128** are exposed, and the mold compound **120** may cover the top surfaces of the package **128** in an embodiment. For one embodiment, the package **128** has a nominal thickness (i.e., z-height) of approximately 420 μm .

The package **100** can be designed to have a nominal thickness (i.e., z-height) of approximately 869 μm to 915 μm and a die-to-substrate ratio that is equal to or greater than 0.85. Furthermore, the use of the pillars **114A-B**, the interposer **106**, and the solder bumps **112** can assist with preventing or minimizing warpage of the package **100**.

With regard now to FIG. 1B, which is a cross-sectional illustration of an ultra-thin, hyper-density semiconductor package **150** that includes a package-on-package (PoP) architecture according one or more embodiments. The package **150** shown in FIG. 1B includes many of the same components described above in connection with the package **100** shown in FIG. 1A. For brevity, only the differences

5

between the package **150** and the package **100** are described below in connection with FIG. **1B**.

One difference between the package **150** and the package **100** is that the package **150** includes pillars **154** (instead of the pillars **114** described above in connection with FIG. **1A**). The pillars **154**, in some embodiments, are designed without solder caps (e.g., the solder caps **138** shown in FIG. **1A**, etc.). In these embodiments, top surfaces of the pillars **154** are exposed through a grinding/polishing process (e.g., chemical mechanical polishing/planarization (CMP) techniques, any other suitable technique, etc.). Consequently, and for these embodiments, no solder caps are required on top of the pillars **154**. As a result, the interposer **106** can be coupled directly to the exposed top surfaces of the pillars **154**. For example, the interposer **106** can be soldered directly to the exposed top surfaces of the pillars **154**.

FIGS. **2A-2I** are cross-sectional side view illustrations of a method of forming an ultra-thin, hyper-density semiconductor package according one or more embodiments. The method shown in FIGS. **2A-2I** can be used, for example, to form the packages **100** and **150** described above in connection with FIGS. **1A-1B**.

Referring now to FIG. **2A**, a HDP substrate **204** with metal pillars **206** is disposed on a carrier substrate **202**, e.g., a silicon wafer, a glass wafer, a metal carrier etc. An adhesive layer (not shown) may be applied on the carrier substrate **102** prior to application of the HDP substrate **204**. The adhesive layer can be a temporary adhesive, e.g., a polyimide adhesive, a polymeric bonding agent, adhesive tapes, etc. Furthermore, and as shown in FIG. **2A**, metal pillars **206** may be formed on the HDP substrate **104**. For one embodiment, the metal pillars **206** may be formed using lithographically-based techniques as is known in the art. The metal pillars **206** may be formed from copper or any other suitable metal or metal alloy.

Referring now to FIG. **2B**, a component **208** (e.g., an SoC chip, etc.) may be transferred onto the HDP substrate **204**. For one embodiment, the component **208** includes one or more semiconductor dies and/or other electrical components. The component **208** may be attached via any suitable chip attach technology (e.g., thermo-compression bonding (TCB) technology, etc.). For one embodiment, the component **208** is attached to the substrate **204** via FLIs **226** and an epoxy material **205**. The epoxy material **205** may also be used to fill gaps between the component **208** and the substrate **204**.

Referring now to FIG. **2C**, the component **208** and the pillars **206** are encapsulated in a first mold compound **210** on the substrate **204**. As used herein, "encapsulating" does not require all surfaces to be encased within a mold compound. For a first example, the lateral sides of the pillars **206** are encased in first mold compound **210**, while the mold compound **210** is not formed over the top surface of the pillars **206**. For a second example, and as illustrated in FIG. **2C**, the lateral and top sides of the component **208** and the pillars **206** are encased in first mold compound **210**. Additional encapsulation operations may be subsequently performed in order to provide chemical and mechanical protection to the top surface of the component **208** and/or the pillars **206**. In some embodiments, the amount of mold compound **210** is controlled to achieve a specified z-height. Alternatively, an amount of the mold compound **210** can be removed after application in order to expose the top and/or lateral surfaces of the component **208** and/or the pillars **206**. As shown in FIG. **2C**, it is not required that the top surfaces of the component **208** and/or the pillars **206** are exposed, and the

6

mold compound **210** may cover the top surfaces of the component **208** and/or the pillars **206** in an embodiment.

Referring now to FIG. **2D**, the mold compound **210** may be removed or etched away to reveal or expose top and/or lateral surfaces of the component **208** and the pillars **206**. For one embodiment, a top surface of the component **208** is exposed via planarization of the mold compound **210** and top surfaces of the pillars are exposed via laser etching techniques. For one embodiment, a fine beam laser may be used to expose and clean top surfaces of the pillars. The laser may also be used to form a shallow hole **236** in each of the pillars **206**.

With regard now to FIG. **2E**, a low temperature solder material (e.g., Sn57 Bi, SAC305, etc.) may be inserted into the shallow holes **236** and reflowed to form solder caps **238**. For one embodiment, the caps **238** are above the mold compound **210** by a predetermined z-height (e.g., approximately 10 μm , etc.). For one embodiment, the low temperature solder may be inserted or applied to the shallow holes **236** via a paste print solder process that involves using a stencil or via an injection molded solder (IMS) process. The solder caps **238** may be planarized and cleaned to achieve the desired z-height.

With regard now to FIGS. **2F(i)-2F(ii)**, an epoxy material **240** with a predetermined thickness (e.g., approximately 25 μm , etc.) and a predetermined thermal conductivity (e.g., approximately 3-5 W/mK, etc.) may be applied on the exposed top surface of the component **208**. The epoxy material **240** can be a paste or a film. When the epoxy material **240** is a paste, it is printed onto the component **208**. When the epoxy material **240** is a film, it is laminated onto the component **208**.

With specific regard again to FIG. **2F(i)**, the pillars **206** are designed with the solder caps **238**, as described above in connection with FIGS. **2D-2E**. Other embodiments, however, are not so limited. For example, and with regard to FIG. **2F(ii)**, pillars **206** may be similar to the pillars **154** described above in connection with FIG. **1B**. In these alternative embodiments, top surfaces of the pillars **206** are exposed through a grinding/polishing process (e.g., chemical mechanical polishing/planarization (CMP) techniques, any other suitable technique, etc.). Consequently, and for these alternative embodiments, no shallow holes **236** and solder caps **238** are required on top of the pillars **206**, and the interposer **242** can be coupled directly to the exposed top surfaces of the pillars **206**. For example, the interposer can be soldered directly to the exposed top surfaces of the pillars **206**.

Moving on to FIG. **2G**, a pitch translation interposer **242** is applied or disposed on exposed top surfaces of the epoxy material **240**, the mold compound **210**, and the solder caps **238**. For one embodiment, the solder caps **238** are reflowed to secure the interposer **242**. With regard now to FIG. **2H**, a package **250** is formed after the carrier substrate **202** is removed, contact pads of the HDP substrate **204** are cleaned, and solder bumps **244** formed from low temperature solder materials (e.g., Sn57Bi, etc.) are attached and reflowed. For one embodiment, the package **250** is designed to have a nominal thickness (i.e., z-height) of approximately 869 μm to 915 μm and a die-to-substrate ratio that is equal to or greater than 0.85. Furthermore, the use of the pillars **206**, the interposer **242**, and/or the solder bumps **244** can assist with preventing or minimizing warpage of the package **100**.

Moving on to FIG. **2I**, another package **246** may optionally be attached to the interposer **242**. The package **246** may include one or more components **286A-B** (e.g., semiconductor dies, other electrical components, etc.) disposed on

one or more layers **294** (e.g., metal layers, dielectric layers, passivation layers, redistribution layers, etc.), where the components **286A-B** and the layer(s) **294** are encapsulated in a second mold compound **292**. The package **246** may be attached to the interposer via any suitable attachment mechanism **296** (e.g., bumps, microbumps, etc.). The attachment mechanism **296** can be formed from solder materials (e.g., low temperature solder materials, etc.).

FIG. **3** is a cross-sectional illustration of an ultra-thin, hyper-density semiconductor package **300** according to another embodiment. For one embodiment, the package **300** is formed using coreless substrate technology, die back metallization, and low temperature solder technology for ball grid array (BGA) metallurgy. As a result, the package **300** can be designed as an ultra-thin, hyper density package that has safeguards against warpage. For a specific embodiment, the package **300** has a nominal thickness (i.e., a z-height) of 425-750 μm .

As shown, the package **300** includes an HDP substrate **304**, a component **308**, first level interconnects (FLIs) **346**, an epoxy layer **305**, a mold compound **310**, a die and mold back metallization layer **342**, and solder bumps **344**. The HDP substrate **304** can comprise at least one hyper density layer and at least one dielectric layer. The HDP substrate **304** can be formed from any suitable material (e.g., silicon, glass, metal, etc.). For one embodiment, the HDP substrate **304** has a nominal thickness (i.e., a z-height) of 150-180 μm .

The component **308** may be coupled to the substrate **304** via FLIs **346** and the epoxy layer **305**. The component **308** can be a semiconductor die or a multiple die configuration. Multiple die configurations can include a variety of passive components, active components, active and passive components, and/or SoCs. Accordingly, a variety of combinations are possible. For one embodiment, the component **308** has a nominal thickness (i.e., a z-height) of 110-300 μm . The component **308** can also be similar to or the same as any of the components described above in connection with FIGS. 1-2I.

The epoxy layer **305** is disposed on the HDP substrate **305** and may be used to fill gaps between the FLIs **346**. The layer **305** can be formed from any suitable epoxy material as is known in the art of semiconductor manufacturing and fabrication (e.g., epoxy resin, phenolic resin, etc.). For one embodiment, the epoxy layer **305** and the FLIs **346** have a combined nominal thickness (i.e., a z-height) of 35 μm .

A mold compound **310** may encapsulate the component **308** and the epoxy layer **305**. For one embodiment, the mold compound **310** has a nominal thickness of 150-180 μm . For one embodiment, top surfaces of the mold compound **310** and the component **308** are co-planar with each other. A die and mold back metallization layer **342** may be disposed on top, exposed surfaces of the mold compound **310** and the component **308**. The metallization layer **342** may be formed from any suitable metal or metal alloy (e.g., copper, etc.) and may include one or more metal layers (e.g., an adhesion layer, etc.). For one embodiment, the metallization layer **342** has a nominal thickness (i.e., a z-height) of 30-100 μm .

The package **300** also includes solder bumps **344** formed on a bottom side of the HDP substrate **304**. The bumps **344** can be formed from any suitable solder materials (e.g., low temperature solder materials, etc.). For a specific embodiment, the bumps **344** are formed from Sn57Bi. The bumps **344** may be designed to have a minimum second level interconnect (SLI) pitch of 0.35. For one embodiment, the bumps **344** have a nominal thickness (i.e., a z-height) of 100-150 μm .

FIG. **4** is a cross-sectional illustration of an ultra-thin, hyper-density semiconductor package **400** according to another embodiment. The package **400** includes many of the same components as the package **300**, which is described above in connection with FIG. **3**. For brevity, only the differences between the package **400** and the package **300** are described below in connection with FIG. **4**.

One difference between the package **300** and the package **400** is that the package **400** includes multiple components **308** and **318**. Each of the components **308** and **318** can be semiconductor dies. Each of the components **308** and **318** can include one or more active and/or passive electronic device components—e.g., transistors, memories, capacitors, resistors, optoelectronic devices, switches, interconnects, and any other electronic device components. For one embodiment, at least one of the components **308** and **318** includes a memory, a processor, a Platform Controller Hub (PCH), a Peripheral Component Interconnect (PCI), a Graphics Processing Unit (GPU), an on-chip system fabric, a network interface controller, a stacked component, a non-stacked component, a ball grid array (BGA) package, any other electronic component, or any combination thereof. For one embodiment, each of the components **308** and **318** has a nominal thickness (i.e., a z-height) of 110-300 μm . Furthermore, the z-heights of the components **308** and **318** may or may not be the same.

FIGS. 5A-5I are cross-sectional side view illustrations of a method of forming an ultra-thin, hyper-density semiconductor package according to another embodiment. For one embodiment, the method shown in FIGS. 5A-5I can be used to fabricate packages similar to the packages **300** and **400** described above in connection with FIGS. 3-4.

With regard now to FIG. 5A, an HDP substrate **504** may be formed or applied to a blank panel **502** (which can also be referred to as a detach core **502**). This blank panel **502** may be a peelable core, and may be constructed with various materials, such as copper (Cu), or one or more other suitable materials, metals, or metal alloys. For example, a blank panel **502** may include several layers of epoxy resin disposed between layers of copper.

Moving on to FIG. 5B, an epoxy layer **506** may be applied on the HDP substrate **504**. The epoxy layer **506** may be formed any suitable epoxy resin or composite that is in a paste form or film form. Examples of materials used to form the epoxy layer include, but are not limited to, an amine epoxy, imidazole epoxy, a phenolic epoxy, and an anhydride epoxy. When the material used to form the epoxy layer **506** is a paste, it is printed onto the substrate **504**. When the material used to form the epoxy layer **506** is a film, it is laminated onto the substrate **504**.

Referring now to FIG. 5C, one or more components **508** (e.g., an SoC chip, a central processing unit (CPU), a platform controller hub (PCH), a power management integrated circuit (PMIC), etc.) may be transferred onto the HDP substrate **504**. For one embodiment, the component(s) **508** include one or more semiconductor dies and/or other electrical components. The component(s) **508** may be attached via any suitable chip attach technology (e.g., thermo-compression bonding (TCB) technology, etc.). For one embodiment, the component(s) **508** are attached to the substrate **504** via FLIs **546** and the epoxy layer **506**. The epoxy material **506** may also be used to fill gaps between the component(s) **508** and the substrate **504**. In addition, and with regard again to FIG. 5C, one or more additional structures (not shown in FIGS. 5A-5I) may be formed on substrate **504**. For one embodiment, the additional structure(s) may be included to assist with propagating signals within the package formed

using the method shown in FIGS. 5A-5I. The additional structure(s) include, but are not limited to one or more metal pillars. These metal pillars may be formed using lithographically-based techniques as is known in the art. The metal pillars may be formed from copper or any other suitable metal or metal alloy.

Moving on FIG. 5D, a mold compound 510 is used to encapsulate the component(s) 508 and/or any other additional structure(s) on the substrate 504 (e.g., pillars, etc.). As used herein, "encapsulating" does not require all surfaces to be encased within a mold compound. Additional encapsulation operations may be subsequently performed in order to provide chemical and mechanical protection to the top surface(s) of the component(s) 508 and/or any other additional structure(s) on the substrate 504 (e.g., pillars, etc.). In some embodiments, the amount of mold compound 510 is controlled to achieve a specified z-height. In the specific embodiment illustrated in FIG. 5D, only component(s) 508 are shown, so only component(s) 508 are encapsulated in the mold compound 510.

Referring now to FIG. 5E, the mold compound 510 may be removed or etched away via any suitable technique to reveal or expose top and/or lateral surfaces of the component(s) 508 and/or at least one of the additional structure(s) on the substrate 504. In the specific embodiment illustrated in FIG. 5E, only component(s) 508 are shown, so the mold compound 510 is removed or etched away to reveal or expose top and/or lateral surfaces of the component(s) 508. For a specific embodiment, planarization of the mold compound 510 is performed until top surface(s) of the component(s) 508 and/or at least one of the additional structure(s) on the substrate 504 are revealed or exposed. In the specific embodiment illustrated in FIG. 5E, only component(s) 508 are shown, so planarization of the mold compound 510 is performed until top surface(s) of the component(s) 508 are revealed or exposed. For one embodiment, exposed top surfaces of the component(s) 508 and top surfaces of the mold compound 510 are co-planar with each other. For one embodiment, exposed top surfaces of the component(s) 508, top surface(s) of at least one additional structure on the substrate 504 that is adjacent to the component(s) 508 (e.g., pillars, etc.), and top surfaces of the mold compound 510 are co-planar with each other.

Referring now to FIG. 5F, the blank panel 502 may be removed or etched away to reveal or expose a bottom surface of the substrate 504. Any suitable removal or etching technique may be used.

With regard now to FIG. 5G, one or more metals layers 512 may be applied on exposed top surfaces of mold compound 510 and the component(s) 508. For one embodiment, the one or more metal layers 512 are applied via sputtering, electroplating, depositing, or any other suitable technique. The one or more metal layers 512 may comprise copper, titanium, or any other suitable metal or metal alloy. For one embodiment, the one or more metal layers 512 include an adhesion layer.

Moving on to FIG. 5H, a die and mold back metallization layer 514 is formed on the one or more metal layers 512. For one embodiment, the layer 514 is formed by electroplating a metal or metal alloy (e.g., copper, etc.) onto the layer(s) 512. For this embodiment, the layer 514 has a nominal thickness (i.e., a z-height) of approximately 30-50 μm . For another embodiment, the layer 514 is formed by printing sinterable bonding material onto the layer(s) 512. For this embodiment, the layer 514 has a nominal thickness (i.e., a z-height) of approximately 50-100 μm . The sinterable bonding material can have a low temperature range (e.g., 150° C.

to 200° C.). The sinterable bonding material can be formed from copper, silver, a copper-silver alloy, or any other suitable metal or metal alloy. For yet another embodiment, the layer 514 is formed by laminating a metal or metal alloy (e.g., copper, etc.) onto the layer(s) 512. For this embodiment, the layer 514 has a nominal thickness (i.e., a z-height) of approximately 30-100 μm . The laminated metal or metal alloy used to form the layer 514 may be black oxide treated and may exhibit a thermal conductivity that is approximately 20 W/mK.

With regard again to FIGS. 5G and 5H, in some embodiments, the metal layers 512 and 514 include metal (stiffener) structures that are electroplated or deposited on the exposed top surfaces of mold compound 510 and the component(s) 508 to provide warpage control. In other embodiments, the layers 512 and 514 include one or more foils (e.g., copper foils, black oxide treated copper foils, any other foils formed from suitable metals or metal alloys, etc.). In these embodiments, the one or more foils may be attached with an adhesive on the exposed top surfaces of mold compound 510 and the component(s) 508 to provide warpage control.

With regard now to FIG. 5I, a package 575 is formed after formation of the layers 512 and 514, contact pads of the HDP substrate 504 are cleaned, and solder bumps 544 formed from low temperature solder materials (e.g., Sn57Bi, etc.) are attached and reflowed. For one embodiment, the package 575 is designed to have a nominal thickness (i.e., z-height) of approximately 425 μm to 750 μm . For one embodiment, the package 575 is designed to have a die-to-substrate ratio that is equal to or greater than 0.70. Furthermore, the use of the layers 512 and 514, the epoxy layer 506, and the solder bumps 544 can assist with preventing or minimizing warpage of the package 575.

Although not shown in FIGS. 5A-5I, the method used to form the package 575 may, in some embodiments, include forming or disposing one or more additional structures that are adjacent to the component(s) 508 and encapsulated in the mold compound 510. The additional structure(s) may include, but are not limited to, pillars (e.g., pillars formed from metal, metal alloys, and/or any other suitable conductive material, etc.). Furthermore, for some embodiments, the layers 512 and 514 may include one or more structures that assist with propagating signals within the package 575 (e.g., vias, pads, traces, redistribution layers, etc.).

FIG. 6 is a cross-sectional illustration of an ultra-thin, hyper-density semiconductor package 600 according one or more embodiments. The package 600 includes solder bumps 612, which may be low temperature solder bumps formed from tin (Sn) or tin alloys (e.g., Sn—Al alloys, Sn—In alloys, Sn—Bi alloys, etc.). For one embodiment, the solder bumps 612 are formed from an Sn-57 Bi-1 alloy.

The package 600 also includes a high density (HDP) ultra-thin substrate 608 onto which the solder bumps 612 are formed. The HDP substrate 608 may be formed from any suitable material (silicon, glass, metal, etc.). For one embodiment, the substrate 608 has a nominal thickness (i.e., z-height) of approximately 66 μm to 70 μm . For one embodiment, a top side of the substrate 608 has metal pillars 614 (e.g., copper pillars, etc.) formed thereon. For one embodiment, the metal pillars 614 have a maximum nominal thickness (i.e., z-height) that is approximately 150 μm .

The package 600 also includes a component 610. For one embodiment, the component 610 can include one or more of a system-on-chip (SoC), a central processing unit (CPU) component, a memory, a processor, a Platform Controller Hub (PCH), a Peripheral Component Interconnect (PCI), a Graphics Processing Unit (GPU), an on-chip system fabric,

11

a network interface controller, a stacked component, a non-stacked component, a ball grid array (BGA) package, any other electronic component, or any combination thereof. The component **610** may include one or more semiconductor dies mounted on the HDP substrate **608**. The dies of the component **610** can be attached to the HDP substrate **608** according to a variety of suitable configurations including, a flip-chip configuration or other configurations such as wire bonding and the like. In the flip-chip configuration, an active side of the dies **602** is attached to a surface of the substrate **608** using interconnect structures such as bumps or pillars. Examples of such interconnect structures include, but are not limited to, Cu bumps, any type of low-lead or lead-free solder bumps, tin-copper bumps, Cu pillars, combinations thereof, or the like. The active side of the dies in the component **610** may have one or more transistor devices formed thereon. Each of the dies in the component **610** may represent a discrete chip. The dies in the component **610** may, include, or be a part of a processor, memory, or application specific integrated circuit (ASIC).

As shown in FIG. 6, the component **610** may be coupled to the substrate **608** via the first level interconnects (FLIs) **626** and an epoxy layer **624**. The FLIs **626** can have a nominal thickness (i.e., a z-height, etc.) of approximately 35 μm . The epoxy layer **624** can have a nominal thickness (i.e., a z-height, etc.) of approximately 25 μm . For some embodiments, a substrate pad may have nominal thickness (i.e., a z-height, etc.) of 40 μm . For some embodiments, a pad-trace component may have nominal thickness (i.e., a z-height, etc.) of 10 μm .

The component **610** and the pillars **614** may be encapsulated in a first mold compound **622**. The component **610** may be a monolithic package (e.g., a monolithic SoC, etc.). Furthermore, the component **610** may be designed to have a nominal thickness (i.e., z-height) of approximately 125 μm .

For one embodiment, top surfaces of the pillars **614**, the component **610**, and the mold compound **622** are co-planar with each other. The co-planar top surfaces may be achieved by grinding/polishing top surfaces of the pillars **614**, the component **610**, and the mold compound **622**.

The package **600** also includes one or more layers **642**. The layer(s) **642** can include a buildup layer (e.g., Ajinomoto Buildup Film (ABF), liquid crystal polymer, benzocyclobutene (BCB), polyimide, prepreg (a weaved fiber network "preimpregnated" into an epoxy matrix), epoxy, a combination thereof, etc.). The layer(s) **642** may include a seed layer formed from conductive materials (e.g., copper, etc.) that is on the buildup layer. The seed layer can be deposited using one of conductive layer deposition techniques, e.g., electroless plating, electroplating, sputtering, chemical vapor deposition (CVD), metalorganic chemical vapor deposition (MOCVD), atomic layer deposition (ALD), or any other conductive layer deposition technique known to one of ordinary skill in the art of electronic device manufacturing.

The layer(s) **642** may also include a photoresist layer that is deposited using a dry film process on the seed layer. In another embodiment, the photoresist layer is deposited by application of a solution using for example, a spin-coating, a slit-coating, a spray-coating, or any other coating technique, or any other photoresist depositing techniques known to one of ordinary skill in the art of electronic device manufacturing. For one embodiment, a thickness of the photoresist layer is determined by the sum of the pad height and the via height. The photoresist layer may be patterned to form metal features on exposed and co-planar surfaces of the pillars **614**, the mold compound **622**, and the component

12

610. Generally, a semi-additive metallization process involves forming a photoresist mask that defines the regions of a substrate on which metal features are formed later on in a process. These metal features includes vias and traces **644**, **645**. The vias and traces **645** may be in contact with one or more pads of the component **610**. For one embodiment, the layer(s) **642** have a nominal thickness (i.e., z-height) of approximately 10 μm . For one embodiment, the layer(s) **642** include one or more redistribution layers.

The package **600** can be designed to have one or more of the following: (i) a nominal thickness (i.e., z-height) of approximately 285 μm to 365 μm ; and (ii) a die-to-substrate ratio that is equal to or greater than 0.85. Furthermore, the pillars **614** and the solder bumps **112** in the package **600** can assist with preventing or minimizing warpage of the package **600**.

FIGS. 7A-7G are cross-sectional side view illustrations of a method of forming an ultra-thin, hyper-density semiconductor package according one or more embodiments. The method shown in FIGS. 7A-7G can be used, for example, to form the package **600** described above in connection with FIG. 6.

Referring now to FIG. 7A, a HDP substrate **704** with metal pillars **706** is disposed on a carrier substrate **702**, e.g., a silicon wafer, a glass wafer, a metal carrier etc. An adhesive layer (not shown) may be applied on the carrier substrate **702** prior to application of the HDP substrate **704**. The adhesive layer can be a temporary adhesive, e.g., a polyimide adhesive, a polymeric bonding agent, adhesive tapes, etc. Furthermore, and as shown in FIG. 7A, metal pillars **706** may be formed on the HDP substrate **704**. For one embodiment, the metal pillars **706** may be formed using lithographically-based techniques as is known in the art. The metal pillars **706** may be formed from copper or any other suitable metal or metal alloy.

Referring now to FIG. 7B, a component **708** (e.g., an SoC chip, etc.) may be transferred onto the HDP substrate **704**. For one embodiment, the component **708** includes one or more semiconductor dies and/or other electrical components. The component **708** may be attached via any suitable chip attach technology (e.g., thermo-compression bonding (TCB) technology, etc.). For one embodiment, the component **708** is attached to the substrate **704** via FLIs **726** and an epoxy material **705**. The epoxy material **705** may also be used to fill gaps between the component **708** and the substrate **704**.

Referring now to FIG. 7C, the component **708** and the pillars **706** are encapsulated in a first mold compound **710** on the substrate **704**. As used herein, "encapsulating" does not require all surfaces to be encased within a mold compound. For a first example, the lateral sides of the pillars **706** are encased in first mold compound **710**, while the mold compound **710** is not formed over the top surfaces of the pillars **706**. For a second example, and as illustrated in FIG. 7C, the lateral and top sides of the component **708** and the pillars **706** are encased in first mold compound **710**. Other examples are possible. Additional encapsulation operations may be subsequently performed in order to provide chemical and mechanical protection to the top surface(s) of the component **708** and/or the pillars **706**. In some embodiments, the amount of mold compound **710** is controlled to achieve a specified z-height. Alternatively, an amount of the mold compound **710** can be removed after application in order to expose the top and/or lateral surfaces of the component **708** and/or the pillars **706**. As shown in FIG. 7C, it is not required that the top surfaces of the component **708** and/or the pillars **706** are exposed, and the mold

13

compound **710** may cover the top surfaces of the component **708** and/or the pillars **706** in an embodiment.

Referring now to FIG. 7D, the mold compound **710** may be removed or etched away via any suitable technique to reveal or expose top and/or lateral surfaces of the component **708** and the pillars **706**. For one embodiment, top surface(s) of the component **708** is exposed via planarization of the mold compound **710**. For one embodiment, top surfaces of the pillars **706** are exposed via laser etching techniques. For one embodiment, a fine beam laser may be used to expose and clean top surfaces of the pillars **706**. For one embodiment, and as shown in FIG. 7D, top surfaces of the pillars **706**, the component **708**, and the mold compound **710** are exposed and co-planar with each other. The exposed and co-planar top surfaces of the pillars **706**, the component **708**, and the mold compound **710** may be achieved through a grinding/polishing process (e.g., chemical mechanical polishing/planarization (CMP) techniques, any other suitable technique, etc.).

Moving on to FIG. 7E, the carrier substrate **702** is removed or etched away. Any suitable removal or etching technique may be used.

With regard now to FIG. 7F, one or more layers **742** are disposed on the exposed and co-planar top surfaces of the pillars **706**, the component **708**, and the mold compound **710**. The layer(s) **742** can include a buildup layer (e.g., Ajinomoto Buildup Film (ABF), liquid crystal polymer, benzocyclobutene (BCB), polyimide, prepreg (a weaved fiber network “preimpregnated” into an epoxy matrix), epoxy, a combination thereof, etc.). The layer(s) **742** may include a seed layer formed from conductive materials (e.g., copper, etc.) that is on the buildup layer. The seed layer can be deposited using one of conductive layer deposition techniques, e.g., electroless plating, electroplating, sputtering, chemical vapor deposition (CVD), metalorganic chemical vapor deposition (MOCVD), atomic layer deposition (ALD), or any other conductive layer deposition technique known to one of ordinary skill in the art of electronic device manufacturing.

The layer(s) **742** may also include a photoresist layer that is deposited using a dry film process on the seed layer. In another embodiment, the photoresist layer is deposited on the seed layer by application of a solution using for example, a spin-coating, a slit-coating, a spray-coating, or any other coating technique, or any other photoresist depositing techniques known to one of ordinary skill in the art of electronic device manufacturing. For one embodiment, a thickness of the photoresist layer is determined by the sum of the pad height and the via height. The photoresist layer may be patterned to form metal features on exposed and co-planar surfaces of the pillars **704**, the mold compound **710**, and the component **708**. Generally, a semi-additive metallization process involves forming a photoresist mask that defines the regions of a substrate on which metal features are formed later on in a process. These metal features includes vias and traces **744**, **745**. The vias and traces **745** may be in contact with one or more pads of the component **708**. For one embodiment, the layer(s) **742** include one or more redistribution layers.

Moving on to FIG. 7G, a package **700** is formed after layer(s) **742** are formed, contact pads of the HDP substrate **704** are cleaned, and solder bumps **746** formed from low temperature solder materials (e.g., Sn57Bi, etc.) are attached and reflowed. For one embodiment, the package **700** can be designed to have one or more of: (i) a nominal thickness (i.e., z-height) of approximately 285 μm to 365 μm ; and (ii) a die-to-substrate ratio that is equal to or greater than 0.84.

14

Furthermore, the pillars **706** and the solder bumps **746** in the package **700** can assist with preventing or minimizing warpage of the package **700**.

FIG. 8 illustrates a schematic of computer system **800** according to an embodiment. The computer system **800** (also referred to as an electronic system **800**) can include a semiconductor package in accord with any of the embodiments and their equivalents as set forth in this disclosure. The computer system **800** may be a mobile device such as a netbook computer. The computer system **800** may be a mobile device such as a wireless smart phone. The computer system **800** may be a desktop computer. The computer system **800** may be a hand-held reader. The computer system **800** may be a server system. The computer system **800** may be a supercomputer or high-performance computing system.

The electronic system **800** can be a computer system that includes a system bus **820** to electrically couple the various components of the electronic system **800**. The system bus **820** is a single bus or any combination of busses according to various embodiments. The electronic system **800** includes a voltage source **830** that provides power to the integrated circuit **810**. For one embodiment, the voltage source **830** supplies current to the integrated circuit **810** through the system bus **820**.

The integrated circuit **810** is electrically coupled to the system bus **820** and includes any circuit, or combination of circuits according to an embodiment. For an embodiment, the integrated circuit **810** includes a processor **812** that can be of any type. As used herein, the processor **812** may mean any type of circuit such as, but not limited to, a microprocessor, a microcontroller, a graphics processor, a digital signal processor, or another processor. For an embodiment, the processor **812** includes, or is coupled with, a semiconductor package in accord with any of the embodiments and their equivalents, as described in the foregoing specification. For an embodiment, SRAM embodiments are found in memory caches of the processor. Other types of circuits that can be included in the integrated circuit **810** are a custom circuit or an application-specific integrated circuit (ASIC), such as a communications circuit **814** for use in wireless devices such as cellular telephones, smart phones, pagers, portable computers, two-way radios, and similar electronic systems, or a communications circuit for servers. For an embodiment, the integrated circuit **810** includes on-die memory **816** such as static random-access memory (SRAM). For an embodiment, the integrated circuit **810** includes embedded on-die memory **816** such as embedded dynamic random-access memory (eDRAM). For one embodiment, the on-die memory **816** may be packaged with a process that is in accord with any of the embodiments and their equivalents, as described in the foregoing specification.

The integrated circuit **810** may be complemented with a subsequent integrated circuit **811**. Useful embodiments include a dual processor **813** and a dual communications circuit **815** and dual on-die memory **817** such as SRAM. For an embodiment, the dual integrated circuit **810** includes embedded on-die memory **817** such as eDRAM.

For an embodiment, the electronic system **800** also includes an external memory **840** that in turn may include one or more memory elements suitable to the particular application, such as a main memory **842** in the form of RAM, one or more hard drives **844**, and/or one or more drives that handle removable media **846**, such as diskettes, compact disks (CDs), digital variable disks (DVDs), flash memory drives, and other removable media known in the

15

art. The external memory **840** may also be embedded memory **848** such as the first die in a die stack, according to an embodiment.

For an embodiment, the electronic system **800** also includes a display device **850** and an audio output **860**. For an embodiment, the electronic system **800** includes an input device such as a controller **870** that may be a keyboard, mouse, trackball, game controller, microphone, voice-recognition device, or any other input device that inputs information into the electronic system **800**. For an embodiment, an input device **870** is a camera. For an embodiment, an input device **870** is a digital sound recorder. For an embodiment, an input device **870** is a camera and a digital sound recorder.

At least one of the integrated circuits **810** or **811** can be implemented in a number of different embodiments, including a semiconductor package that is in accord with one or more of the embodiments described in the foregoing specification and their art-recognized equivalents, an electronic system, a computer system, one or more methods of fabricating an integrated circuit, and one or more methods of fabricating an electronic assembly that includes a semiconductor package that is in accord with one or more of the embodiments described in the foregoing specification and their art-recognized equivalents. The elements, materials, geometries, dimensions, and sequence of operations described in one or more embodiments described herein can all be varied to suit particular I/O coupling requirements including array contact count, array contact configuration for a microelectronic die embedded in a processor mounting substrate. A foundation substrate may be included, as represented by the dashed line of FIG. **8**. Passive devices may also be included, as is also depicted in FIG. **8**.

Reference throughout this specification to “one embodiment,” “an embodiment,” “another embodiment” and their variations means that a particular feature, structure, configuration, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrase “in one embodiment,” “in an embodiment,” “in another embodiment,” or their variations in various places throughout this specification are not necessarily referring to the same embodiment. Furthermore, the particular features, structures, configurations, or characteristics may be combined in any suitable manner in one or more embodiments.

The terms “over,” “to,” “between,” “onto,” and “on” as used in the foregoing specification refer to a relative position of one layer with respect to other layers. One layer “over” or “on” another layer or bonded “to” or in “contact” with another layer may be directly in contact with the other layer or may have one or more intervening layers. One layer “between” layers may be directly in contact with the layers or may have one or more intervening layers.

The descriptions provided above in connection with one or more of the embodiments described herein (e.g., descriptions of processes provided in connection with FIGS. **1A-8**, etc.) may also be used for other types of IC packages and mixed logic-memory package stacks. In addition, the processing sequences may be compatible with both wafer level packages (WLP), and integration with surface mount substrates such as LGA, QFN, and ceramic substrates.

In the foregoing specification, abstract, and/or Figures, numerous specific details are set forth, such as specific materials and processing operations, in order to provide a thorough understanding of embodiments described herein. It will, however, be evident that any of the embodiments described herein may be practiced without these specific

16

details. In other instances, well-known features, such as the integrated circuitry of semi conductive dies, are not described in detail in order to not unnecessarily obscure embodiments described herein. Furthermore, it is to be understood that the various embodiments shown in the Figures and described in connection with the Figures are illustrative representations and are not necessarily drawn to scale. Thus, various modifications and/or changes may be made without departing from the broader spirit and scope of the embodiments described in connection with the foregoing specification, abstract, and/or Figures.

Embodiments described herein include a semiconductor package, comprising: a substrate; a plurality of metal pillars formed on a top surface of the substrate; a semiconductor component disposed on the substrate, the semiconductor component comprising one or more dies; a mold compound encapsulating the plurality of metal pillars and the semiconductor component; an interposer coupled to the plurality of metal pillars; and a plurality of solder bumps formed on a bottom surface of the substrate.

Additional embodiments described herein include a semiconductor package, wherein each metal pillar includes a solder cap that is partially encapsulated by the mold compound and wherein the interposer is coupled to the plurality of metal pillars via the solder caps.

Additional embodiments described herein include a semiconductor package, wherein a z-height of the package is less than or equal to 1 mm. Additional embodiments described herein include a semiconductor package, wherein a z-height of the package is approximately 869 μm to 915 μm .

Additional embodiments described herein include a semiconductor package, wherein a die-to-package ratio for the package is equal to or greater than 0.7.

Additional embodiments described herein include a semiconductor package, wherein the plurality of metal pillars has a pitch that is greater than or equal to 150 μm .

Additional embodiments described herein include a semiconductor package, wherein a second package is disposed on interposer.

Additional embodiments described herein include a semiconductor package, wherein a second package is disposed on interposer via an attachment mechanism.

Additional embodiments described herein include a semiconductor package, wherein a z-height of the substrate is approximately 66 μm .

Additional embodiments described herein include a semiconductor package, wherein a z-height of the semiconductor component is approximately 125 μm .

Additional embodiments described herein include a semiconductor package, wherein a z-height of the interposer is approximately 63 μm .

Embodiments described herein include a semiconductor package, comprising: a substrate; an epoxy layer disposed on a top surface of the substrate; a semiconductor component disposed on the epoxy layer; a mold compound encapsulating the epoxy layer and the semiconductor component, wherein top surfaces of the mold compound and the semiconductor component are co-planar with each other; a metallization layer formed on the top surfaces of the mold compound and the semiconductor component; and a plurality of solder bumps formed on a bottom surface of the substrate.

Additional embodiments described herein include a semiconductor package, wherein the semiconductor component comprises one or more semiconductor dies.

Additional embodiments described herein include a semiconductor package, wherein a z-height of the package is less

than or equal to 1 mm. Additional embodiments described herein include a semiconductor package, wherein a z-height of the package is approximately 425 μm to 750 μm .

Additional embodiments described herein include a semiconductor package, wherein a die-to-package ratio for the package is equal to or greater than 0.7.

Additional embodiments described herein include a semiconductor package, wherein the epoxy layer has a z-height that is approximately 35 μm .

Additional embodiments described herein include a semiconductor package, wherein the metallization layer has a z-height that is approximately 30 μm to 100 μm .

Additional embodiments described herein include a semiconductor package, wherein a z-height of the plurality of solder bumps is approximately 100 μm to 150 μm .

Additional embodiments described herein include a semiconductor package, wherein a z-height of the semiconductor component is approximately 110 μm to 300 μm . Additional embodiments described herein include a semiconductor package, wherein a z-height of the substrate is approximately 150 μm to 180 μm .

Embodiments described herein include a method of forming a semiconductor package, comprising: forming a plurality of metal pillars on a top surface of a substrate; disposing a semiconductor component on the top surface of the substrate; encapsulating the plurality of metal pillars and the semiconductor component in a mold compound; coupling an interposer to one or more of the plurality of metal pillars and the semiconductor component; and forming a plurality of solder bumps on a bottom surface of the substrate.

Additional embodiments described herein include a method, further comprising disposing a second package on the interposer.

Additional embodiments described herein include a semiconductor package, wherein a second package is disposed on interposer via an attachment mechanism.

Additional embodiments described herein include a method, wherein a z-height of the package is approximately 869 μm to 915 μm . Additional embodiments described herein include a method, further comprising: forming, in each metal pillar, a solder cap that is partially encapsulated by the mold compound, wherein the interposer is coupled to the plurality of metal pillars via the solder caps.

Embodiments described herein include a method of forming a semiconductor package, comprising: disposing an epoxy layer on a top surface of a substrate; disposing a semiconductor component on the epoxy layer; encapsulating the epoxy layer and the semiconductor component in a mold compound, wherein top surfaces of the mold compound and the semiconductor component are co-planar with each other; forming a metallization layer on the top surfaces of the mold compound and the semiconductor component; and forming a plurality of solder bumps on a bottom surface of the substrate.

Additional embodiments described herein include a method, wherein a z-height of the package is approximately 425 μm to 750 μm .

Embodiments described herein include a semiconductor package, comprising: a substrate; a plurality of metal pillars formed on a top surface of the substrate; a semiconductor component disposed on the substrate, the semiconductor component comprising one or more dies; a mold compound encapsulating the plurality of metal pillars and the semiconductor component, wherein top surfaces of the mold compound, the plurality of metal pillars, and the semiconductor component are co-planar with each other; and one or more

layers disposed on the top surfaces, the one or more layers comprising one or more vias and traces.

Additional embodiments include a semiconductor package, wherein a z-height of the package is approximately 285 μm to 365 μm .

Additional embodiments include a semiconductor package, wherein a die-to-substrate ratio of the package is equal to or greater than 0.84.

Embodiments described herein include a method of forming a semiconductor package, comprising: forming a plurality of metal pillars on a top surface of a substrate; disposing a semiconductor component on the substrate, the semiconductor component comprising one or more dies; encapsulating the plurality of metal pillars and the semiconductor component in a mold compound, wherein top surfaces of the mold compound, the plurality of metal pillars, and the semiconductor component are co-planar with each other; and disposing one or more layers on the top surfaces, the one or more layers comprising one or more vias and traces.

Additional embodiments include a method, wherein a z-height of the package is approximately 285 μm to 365 μm .

Additional embodiments include a method, wherein a die-to-substrate ratio of the package is equal to or greater than 0.84.

In the description, drawings, and claims provided herein, the use of “at least one of A, B, and C”, “at least one of A, B, or C”, “one or more of A, B, or C”, or “one or more of A, B, and C” is intended encompass: (i) A alone; (ii) B alone; (iii) C alone; (iv) A and B together; (v) A and C together; (vi) B and C together; or (vii) A, B, and C together. Furthermore, the use of “A, B, and/or C” is intended encompass: (i) A alone; (ii) B alone; (iii) C alone; (iv) A and B together; (v) A and C together; (vi) B and C together; or (vii) A, B, and C together. It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For a first example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.” For a second example, the phrase “A and/or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

The terms used in the following claims should not be construed to limit any of the embodiments described in connection with the foregoing specification, abstract, and/or Figures to the specific embodiments set forth in the foregoing specification, abstract, Figures, and/or claims. Rather, the scope of the claims are to be construed in accordance with established doctrines of claim interpretation.

What is claimed is:

1. A semiconductor package, comprising:

- a substrate having a first surface opposite a second surface, the substrate having a first lateral width;
- a component having a first surface opposite a second surface, and a first side and a second side between the first surface and the second surface, the second side opposite the first side, the second surface of the component coupled to the first surface of the substrate by a plurality of interconnects;
- an epoxy layer between the second surface of the component and the first surface of the substrate, the epoxy layer in lateral contact with individual ones of the plurality of interconnects;
- a first pillar comprising copper, the first pillar on the first surface of the substrate, the first pillar having a first side

19

- and a second side, the second side opposite the first side, the second side of the first pillar laterally spaced apart from the first side of the component, and the first pillar extending above the first surface of the component;
- a second pillar comprising copper, the second pillar on the first surface of the substrate, the second pillar having a first side and a second side, the second side opposite the first side, the first side of the second pillar laterally spaced apart from the second side of the component, and the second pillar extending above the first surface of the component;
- a mold compound on the first surface of the substrate, the mold compound in lateral contact with the first side of the first pillar, the second side of the first pillar, the first side of the component, the second side of the component, the first side of the second pillar, and the second side of the second pillar, the mold compound having a second lateral width the same as the first lateral width, and the mold compound having an uppermost surface above the first surface of the component;
- a pitch translation interposer on the mold compound and having a bottommost surface above and in contact with the uppermost surface of the mold compound, the pitch translation interposer in contact with the first pillar and the second pillar, the bottommost surface of the pitch translation interposer vertically spaced apart from the first surface of the component, and the pitch translation interposer having a third lateral width the same as the second lateral width; and
- a plurality of solder balls on the second surface of the substrate.
2. The semiconductor package of claim 1, wherein the mold compound is in contact with the epoxy layer.
3. The semiconductor package of claim 1, wherein the component is a semiconductor die.
4. The semiconductor package of claim 1, wherein the mold compound is not in contact with the first surface of the component.
5. The semiconductor package of claim 1, wherein the pitch translation interposer has features having a first pitch on a first side, and has features having a second pitch on a second side of the pitch translation interposer, the second pitch different from the first pitch.
6. The semiconductor package of claim 1, further comprising:
- a second die above and coupled to the pitch translation interposer.
7. The semiconductor package of claim 1, further comprising:
- a third die above and coupled to the pitch translation interposer, the third die laterally spaced apart from the second die.
8. The semiconductor package of claim 1, further comprising:
- a second epoxy layer between the first surface of the component and the pitch translation interposer.
9. A semiconductor package, comprising:
- a substrate having a first surface opposite a second surface, the substrate having a first side and a second side, the second side laterally opposite the first side;
- a semiconductor die having a first surface opposite a second surface, and a first side and a second side between the first surface and the second surface, the second side opposite the first side, the second surface of the semiconductor die coupled to the first surface of the substrate by a plurality of interconnects;

20

- a layer between the second surface of the semiconductor die and the first surface of the substrate, the layer in lateral contact with individual ones of the plurality of interconnects;
- a first pillar comprising copper, the first pillar on the first surface of the substrate, the first pillar extending above the first surface of the semiconductor die;
- a second pillar comprising copper, the second pillar on the first surface of the substrate, the second pillar extending above the first surface of the semiconductor die;
- a mold compound on the first surface of the substrate, the mold compound in lateral contact with the first pillar, the first side of the semiconductor die, the second side of the semiconductor die, and the second pillar, the mold compound extending from the first side to the second side of the substrate, and the mold compound having an uppermost surface above the first surface of the semiconductor die;
- a connection layer on the mold compound and having a bottommost surface above and in contact with the uppermost surface of the mold compound, the connection layer in contact with the first pillar and the second pillar, the bottommost surface of the connection layer vertically spaced apart from the first surface of the semiconductor die, and the connection layer extending from the first side to the second side of the substrate, wherein the connection layer has features having a first pitch on a first side, and has features having a second pitch on a second side, the second side over the first side, the second pitch different from the first pitch; and
- a plurality of solder balls on the second surface of the substrate.
10. The semiconductor package of claim 9, wherein the mold compound is in contact with the layer, and wherein the mold compound is not in contact with the first surface of the semiconductor die.
11. The semiconductor package of claim 9, further comprising:
- a second die above and coupled to the connection layer, and
- a third die above and coupled to the connection layer, the third die laterally spaced apart from the second die.
12. The semiconductor package of claim 9, further comprising:
- an epoxy layer between the first surface of the semiconductor die and the connection layer.
13. A method of fabricating a semiconductor package, the method comprising:
- providing a substrate having a first surface opposite a second surface, the substrate having a first lateral width;
- coupling a component to the first surface of the substrate by a plurality of interconnects, the component having a first surface opposite a second surface, and a first side and a second side between the first surface and the second surface, the second side opposite the first side;
- forming an epoxy layer between the second surface of the component and the first surface of the substrate, the epoxy layer in lateral contact with individual ones of the plurality of interconnects;
- forming a first pillar comprising copper, the first pillar on the first surface of the substrate, the first pillar having a first side and a second side, the second side opposite the first side, the second side of the first pillar laterally spaced apart from the first side of the component, and the first pillar extending above the first surface of the component;

21

forming a second pillar comprising copper, the second pillar on the first surface of the substrate, the second pillar having a first side and a second side, the second side opposite the first side, the first side of the second pillar laterally spaced apart from the second side of the component, and the second pillar extending above the first surface of the component;

forming a mold compound on the first surface of the substrate, the mold compound in lateral contact with the first side of the first pillar, the second side of the first pillar, the first side of the component, the second side of the component, the first side of the second pillar, and the second side of the second pillar, the mold compound having a second lateral width the same as the first lateral width, and the mold compound having an uppermost surface above the first surface of the component;

forming a pitch translation interposer on the mold compound, the pitch translation interposer having a bottommost surface above and in contact with the uppermost surface of the mold compound, the pitch translation interposer in contact with the first pillar and the second pillar, the bottommost surface of the pitch translation interposer vertically spaced apart from the

22

first surface of the component, and the pitch translation interposer having a third lateral width the same as the second lateral width; and

forming a plurality of solder balls on the second surface of the substrate.

14. The method of claim 13, wherein the mold compound is in contact with the epoxy layer.

15. The method of claim 13, wherein the component is a semiconductor die.

16. The method of claim 13, wherein the mold compound is not in contact with the first surface of the component.

17. The method of claim 13, wherein the pitch translation interposer has features having a first pitch on a first side, and has features having a second pitch on a second side of the pitch translation interposer.

18. The method of claim 13, further comprising: coupling a second die to the pitch translation interposer.

19. The method of claim 13, further comprising: coupling a third die to the pitch translation interposer, the third die laterally spaced apart from the second die.

20. The method of claim 13, further comprising: forming a second epoxy layer between the first surface of the component and the pitch translation interposer.

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